

Photoshop Intermediate: From Practitioner to Pro

Your working notebook for the whole course — all eight modules. Read the key ideas, study the worked examples, do the exercises in the space provided, and check yourself with the quizzes.

8 modules • 23 lessons

Write-in exercises & self-checks

Name: _____

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Advanced Selection & Masking

In the intro you painted masks by hand and let Select Subject find edges.

Now the image does more of the work: luminosity masks built from its own brightness, channel masks built from one color's brightness, Blend-If that hides by tone, and a four-step pipeline for the hardest common cutout. Deliverable: a flawless cutout of flowing hair on a new background.

Lessons **1.1–1.3** (L01–L03)

3 lessons

Session 1 • 3 hours

Name: _____

HOW TO USE THIS WORKBOOK

Each lesson has **Key Ideas**, a **Worked Example** (the steps you'll follow in class), the lesson's **Recipe Card**, **Your Turn** exercises, a **Self-Check**, a **Journal** prompt, and space for **Notes**. Shortcuts are written Windows first, Mac second: **Alt**/**⌘** means Alt on Windows, Option on a Mac. Practice files are in `labs/module-1/practice/`; your own photos work even better.

BEFORE YOU START: YOUR PROFESSIONAL BASELINE

Not a lesson, nothing to deliver: the setup habits the whole course assumes. Tick them off in the Session 1 setup check.

- ◆ ☐ **Know your working space.** **Edit ▶ Color Settings:** my RGB working space says _____.
(Don't change it yet: Module 2 goes deep on this.)
- ◆ ☐ **Work in 16-bit when you can.** **Image ▶ Mode ▶ 16 Bits/Channel** before building masks. It can't add detail an 8-bit file never had, but it stops new banding as you edit.
- ◆ ☐ **Diagnose before you edit.** **Window ▶ Histogram:** look where the tones sit before you touch a slider.
- ◆ ☐ **Calibrate your monitor** if you do color-critical work, and set up shortcuts for the tools you reach for most.
- ◆ ☐ I can open the **Channels** panel (**Window ▶ Channels**). It's today's main stage.

WHEN "IT'S NOT DOING ANYTHING," RUN THE FOUR-POINT CHECK

1. Which panel am I in (Layers or Channels)? 2. Which channel or layer is highlighted? 3. Which thumbnail has the border (image or mask)? 4. Is a selection active (marching ants, or the "not visible" warning)? Nearly every stuck moment in this module is one of these four.

MODULE PROGRESS TRACKER

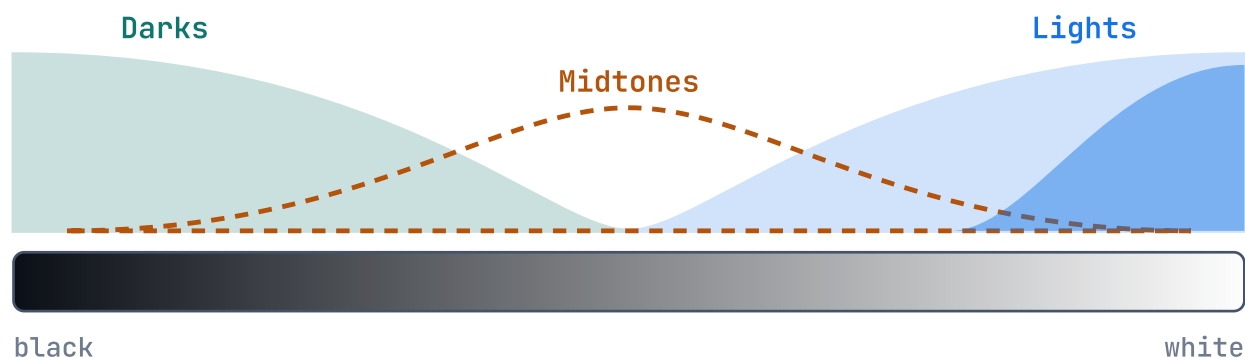
- ◆ ☐ L01 I recovered a sky and lifted a foreground with two luminosity-masked Curves, no visible seam (`luminosity-working.psd`)
- ◆ ☐ L02 I cut out an intricate subject with a channel mask and blended a texture with Blend-If (`channel-working.psd`)
- ◆ ☐ L03 I cut out flowing hair with the four-step pipeline and passed the saturated-background test (`cutout-working.psd` , the Module 1 deliverable)
- ◆ ☐ I wrote a journal entry after each lesson

◆ BY THE END YOU CAN

- ◆ Explain a luminosity mask: a selection built from the image's own luminance
- ◆ Load a **Lights** or **Darks** selection from the channels
- ◆ Adjust highlights and shadows independently, edges self-feathered and invisible
- ◆ Narrow a mask to a tonal band (just the brightest highlights) by **intersecting**

◆ KEY IDEAS

- ◆ **The image paints the mask.** Load a Lights mask and the bright pixels become white (selected, editable) while the dark pixels become black (protected). The mask is a grayscale copy of the photo's luminance.
- ◆ **The edges are perfect.** The mask follows the actual tones, so its transitions are exactly as soft as the light in the scene: no halos, no hand-feathering.
- ◆ **It's self-targeting.** The sky is the bright part, so a Lights mask already contains it. No painting around the horizon.
- ◆ **Invert Lights and you have Darks.** Intersect Lights with itself and the selection slides into the brightest tones only.
- ◆ **Why two curves beat one:** a single S-curve brightens the sky while it lifts the land. Two masked curves move highlights and shadows in *opposite* directions at once.



Overlapping tonal selections. The darker blue wedge is Lights intersected with Lights: only the brightest band.

IN THE CHANNELS PANEL, ON THE RGB THUMBNAIL

DOES

Ctrl/⌘-click

Load Lights (a new selection)

Ctrl/⌘ + Shift-click

Add to the current selection

`Ctrl/⌘ + Alt/⌘-click`

Subtract from the current selection

`Ctrl/⌘ + Shift + Alt/⌘-click`

Intersect: keep only what both share

◆ WORKED EXAMPLE: RESCUE THE LANDSCAPE

- 1 Open a high-contrast photo (or `l01-high-contrast-landscape.jpg`). **Image ▶ Mode ▶ 16 Bits/Channel**, then **Save As** `luminosity-working.psd`.
- 2 **Diagnose, then load Lights.** Read the histogram: highlights piling near the right, shadows near the left. In **Channels**, `Ctrl/⌘-click` the **RGB** thumbnail.
- 3 **Tame the highlights.** With the selection active, add a **Curves** adjustment layer (the selection becomes its mask) and pull the curve down to bring the sky back. Spilling onto the land? Undo the layer, reload Lights, intersect once or twice, then add Curves again.
- 4 **Lift the shadows.** Reload the RGB luminosity selection, invert it (`Ctrl/⌘ + Shift + I`), add another **Curves** layer, and lift the shadows.
- 5 **Check and finish.** Zoom to 100% along the horizon: no seam, no glow. A glow means the masks overlap: reduce one layer's opacity or re-intersect. Optional gentle global Curves on top. Save the layered master.

◆ RECIPE CARD: LUMINOSITY MASK

- 1 Channels panel: `Ctrl/⌘-click` RGB = Lights selection
- 2 Add a Curves layer → the selection becomes its mask
- 3 Too broad? Intersect (`Ctrl/⌘ + Shift + Alt/⌘-click` RGB) to reach extreme tones
- 4 Shadows: reload, invert (`Ctrl/⌘ + Shift + I`), add another Curves
- 5 Check the horizon at 100% for seams; work in 16-bit

Mantra: let the image paint the mask.

◆ YOUR TURN 1: SEE THE MASK THE IMAGE PAINTED

`Alt/⌘-click` your sky Curves layer's mask thumbnail to view the mask itself (click again to return). Sketch it, and describe what happens along the pine trees and the horizon. How long would that edge take to paint by hand?

◆ YOUR TURN 2: GLOW, MIDTONES, AND A HEAD-TO-HEAD

On a portrait or still life (or `101-still-life-window-light.jpg`): (a) load Lights and add a subtle glow by lifting only the brightest highlights. Do the shadows change? (b) Build a **Midtones** mask: save Lights and Darks as channels (**Select ► Save Selection**), then **Select ► All** and **Ctr⌘/⌘ + Alt/⌘**-click each saved channel to subtract it; add contrast to only the mids. (c) On your landscape, compare a luminosity-masked sky darken with a hand-painted one.

Think: why is Lights *minus* Darks not a midtones mask? What does it select instead?

◆ WATCH OUT

Ctrl/⌘-click the **RGB channel** in Channels, not the layer thumbnail in Layers (that loads the whole image). Add Curves **while** the selection is active, or the mask comes out white. "No pixels are more than 50% selected" just means the ants can't show; the selection is still there.

◆ SELF-CHECK

Q1

What is a luminosity mask?

- A) A mask you paint by hand with a big soft brush along the horizon
- B) A selection built from the image's own brightness, saved as a mask
- C) The mask Select Subject makes when it finds the sky
- D) A Levels adjustment applied to the whole image at once

Q2

How do you load a Lights selection?

- A) Select ► Subject, then Select ► Inverse
- B) Alt/⌘-click the photo's thumbnail in the Layers panel
- C) Double-click the Blue channel
- D) Ctrl/⌘-click the RGB thumbnail in the Channels panel

Q3

Your luminosity-masked sky darken is spilling onto the land. What's the lesson's fix?

- A) Undo the layer, reload Lights, and intersect once or twice (Ctrl/⌘ + Shift + Alt/⌘-click RGB) before adding the adjustment
- B) Paint a hard black line along the horizon on the mask
- C) Drop the Curves layer's opacity to 10%
- D) Convert the file to 8-bit so the mask is simpler

Q4

After taming the sky, how do you select the dark land for the second Curves layer?

- A) Ctrl/⌘-click the Blue channel
- B) Intersect Lights with RGB twice more
- C) Reload the RGB luminosity selection, then invert it (Ctrl/⌘ + Shift + I)
- D) Select ► Deselect, then add Curves with no selection

Q5

Why use two luminosity-masked Curves instead of one global S-curve?

- A) Photoshop allows only one unmasked Curves layer per document
- B) A single curve would brighten the sky while it lifts the land; two masked curves move highlights and shadows in opposite directions at once
- C) Masked Curves layers render faster than global ones
- D) An S-curve always clips the highlights

Q6

Your luminosity masks look blocky and banded. What's the most likely cause and fix?

- A) The selection wasn't intersected enough; intersect again
- B) The mask needs a 50 px Gaussian Blur
- C) The Curves layer should be set to the Luminosity blend mode
- D) You're working in 8-bit; convert to 16-bit (Image ► Mode ► 16 Bits/Channel) before building the masks

◆ JOURNAL

Where have you fought a hand-painted mask that a luminosity mask would have solved instantly? Name one past edit you would redo with this technique. Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

◆ BEFORE THE NEXT LESSON

- ◆ Rescue two more high-contrast scenes with Lights + Darks curves.
- ◆ Build and save a Midtones mask: Select All, then subtract your saved Lights and Darks channels.

IF THE MODIFIERS FEEL LIKE FINGER GYMNASTICS

Everyone fumbles the three-key intersect for a day. Put the four-row table above on a sticky note next to your keyboard. By the second landscape, your fingers will know it; by the third, you'll stop seeing skies you can't save.

◆ **NOTES**

◆ BY THE END YOU CAN

- ◆ Read the Red, Green & Blue channels as ready-made grayscale masks
- ◆ Turn the highest-contrast channel into a selection (duplicate, push with Levels)
- ◆ Use **Blend-If** to drop out a layer's lights or darks instantly
- ◆ Pick the right technique for the subject: luminosity, channel, Blend-If, or a blend of all three

◆ KEY IDEAS

- ◆ **Three masks you already have.** Every RGB image is three grayscale images stacked: Red, Green, Blue. Each is a potential mask. You shop for the one with the most subject/background separation.
- ◆ **Skies are blue**, so the Blue channel keeps a sky bright while dark branches stay dark. (For skin, Blue is where blemishes stand out most: same habit, different job.)
- ◆ **Forge, don't paint:** duplicate the winning channel, then push the copy toward black and white with **Image ▶ Adjustments ▶ Levels**. Adjustment layers can't target a channel.
- ◆ **White loads as the selection.** Against a bright sky, the sky loads first, so invert to get the subject.
- ◆ **Blend-If** (Layer Style ▶ Blending Options) hides a layer's pixels by tone in real time. **This Layer** white slider inward drops the layer's lights; black slider inward drops its darks. **Alt/⌘-drag** to split a slider for a soft blend.
- ◆ **When to use which:** Blend-If is instant and non-destructive but tone-global. A channel mask is editable and spatial: you can paint on it afterward.



The whole skill is choosing the right channel; the rest is two clicks and a Levels push.

◆ WORKED EXAMPLE: EXTRACT A TREE

- 1 Open a bare tree, power lines, or other intricate subject against plain bright sky (or `l02-bare-tree-sky.jpg`). 16-bit, **Save As** `channel-working.psd`.
- 2 **Channel-shop.** In Channels, click Red, then Green, then Blue. Pick the one where the subject is darkest and the background brightest.

- 3 **Forge the mask.** Drag the winner onto the New Channel icon. With only the copy selected, **Image ▶ Adjustments ▶ Levels** (**C**tr**l**/**⌘** + **L**): pull the black and white points inward until the sky is white and the tree black, fine twigs still intact. Paint stray gaps with a hard black or white brush on the channel.
- 4 **Load it and cut.** **C**tr**l**/**⌘**-click the copy, invert (**C**tr**l**/**⌘** + **S**h**i**ft + **I**), click **RGB**, go to Layers with the image layer active, and add a layer mask. Drop a new sky behind it (**l02-new-sky.jpg**) to check the edges.
- 5 **A Blend-If finish.** Place a texture (**l02-grunge-texture.jpg**) as a new layer on top. Blending Options: pull the **This Layer** white slider left until only the darker parts remain; **A**l**t**/**⌘**-split for a soft blend. No mask at all.

◆ RECIPE CARD: CHANNEL MASK

- 1 Channels panel: click R / G / B, pick the highest-contrast one
- 2 Duplicate it → Image ▶ Adjustments ▶ Levels to force near black & white
- 3 Clean gaps with a hard brush on the channel
- 4 **C**tr**l**/**⌘**-click to load → invert (the white background loads first) → click RGB → add layer mask
- 5 Fast alternative: Blend-If to drop lights or darks by tone (**A**l**t**/**⌘**-split to feather)

Mantra: shop the channels before you paint.

◆ YOUR TURN 1: CHANNEL-SHOP OUT LOUD

For the tree file, rate each channel's separation from 1 (useless) to 5 (nearly a mask already), and say why.

CHANNEL	SCORE (1-5)	WHAT I SEE
Red		
Green		
Blue		

◆ YOUR TURN 2: BLEND-IF THE OTHER WAY, AND A BUSY BACKGROUND

(a) Place **l02-fireworks-on-black.jpg** (or your own flames, smoke, or fireworks) over **l02-new-sky.jpg** or a night scene, and drop the layer's **shadows** with the This Layer black slider. What happens when you **A**l**t**/**⌘**-split it? (b) Extract something intricate against a *busy* background: channel mask for the easy 90%, then paint and luminosity-mask the rest. (c) Open a portrait's Blue channel: what does it do to skin texture?

Choose the tool: for each subject, write luminosity, channel, Blend-If, or a combination. A blown sky above a bright shirt: _____ · A leafless tree on blue sky: _____ · Smoke on a black background: _____ · Hair on a busy street: _____

◆ WATCH OUT

Run Levels only with the **copy** highlighted, or you'll change the photo's colors. Photo looks gray back in Layers? Click **RGB**. Don't over-push: the finest twigs go white and vanish. For Blending Options, double-click the layer **thumbnail** or empty area, not the name.

◆ SELF-CHECK

Q1

Why can a single color channel work as a mask?

- A) Each of Red, Green and Blue is a full-resolution grayscale image, and a mask is just a grayscale image
- B) Channels are automatically saved as layer masks when you open a file
- C) Only the Blue channel stores brightness information
- D) Channels hold the image's transparency, like a layer mask

Q2

A bare tree against a blue sky: which channel usually gives the best separation?

- A) Red
- B) Green
- C) Blue
- D) The RGB composite, always

Q3

Why duplicate the winning channel before you run Levels on it?

- A) Levels is grayed out on the original channels
- B) A duplicate channel loads as a selection faster
- C) Adjustment layers can target a duplicate channel but not the original
- D) The original channel is part of the photo's color; pushing it would damage the image, while the copy is a safe scratch mask

Q4

You pushed Levels hard and the finest twigs disappeared from the mask. What happened?

- A) The channel was in 16-bit
- B) You over-pushed: the delicate twigs went white; back off until the big areas are solid and the twigs survive
- C) You forgot to invert the selection
- D) Levels always removes detail; use Curves instead

Q5

You Ctrl/Cmd-click your channel copy and the sky is selected, not the tree. Why, and what next?

- A) White loads as the selection; invert (Ctrl/Cmd + Shift + I), click RGB, then add the layer mask on the image layer
- B) The channel is broken; start again from the Red channel
- C) Blend-If is on; turn it off in Blending Options
- D) You loaded the wrong document; reopen the file

Q6

You placed a dark texture on a white background over your scene. Which Blend-If move drops the white?

- A) Drag the Underlying Layer black slider right
- B) Drag the This Layer black slider right
- C) Lower the layer's Fill to 0%
- D) Drag the This Layer white slider left (Alt/⌘-drag to split it for a soft blend)

◆ JOURNAL

Channel masking feels like a cheat code the first time it works. What subject have you given up on cutting out before that you would now attempt? Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

◆ BEFORE THE NEXT LESSON

- ◆ Channel-mask two intricate subjects against sky.
- ◆ Knock a white-background texture into a scene with Blend-If alone.

CHANNELS LOOK INTIMIDATING, THEN BECOME A FAVORITE

The panel full of gray thumbnails is the least friendly-looking place in Photoshop, and within a week it's one of the most useful. You've just added the technique that separates people who "try to select the tree" from people who simply extract it.

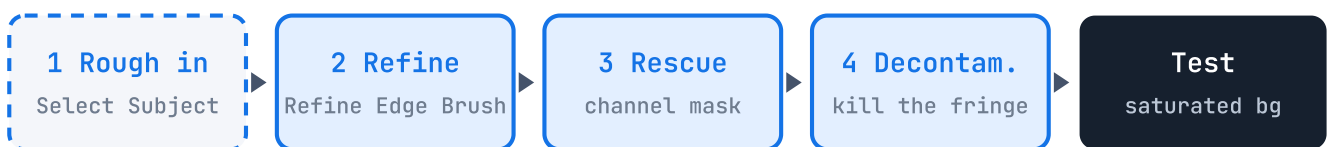
◆ NOTES

◆ BY THE END YOU CAN

- ◆ Run the full **Select & Mask** workflow, tuned for hair
- ◆ Rebuild wispy strands with the **Refine Edge Brush**
- ◆ Rescue what refinement misses with a **channel mask** (Lesson 1.2)
- ◆ Kill the fringe with **Decontaminate Colors** and pressure-test the edge

◆ KEY IDEAS

- ◆ **A pipeline, not a tool.** There is no one button for hair. Each step cleans up what the previous one couldn't.
- ◆ **Rough in** the solid mass (Select Subject, Quick Selection) → **refine** the hairline (Refine Edge Brush) → **rescue** missing strands (channel mask painted into the layer mask) → **decontaminate** the edge (the old background's color lets go).
- ◆ **Refine the edge, not the middle.** The Refine Edge Brush across solid hair turns it translucent and mushy.
- ◆ **Decontaminate last.** It rewrites edge pixels, so it can only output to a new layer or document.
- ◆ **The standard:** fine strands kept, no colored halo against a garish test background, a soft (not razor) edge.



Rough in, refine, rescue, decontaminate, in that order. The garish test background exposes any surviving fringe.

◆ WORKED EXAMPLE: THE HARD CUTOUT

- 1 Open a portrait with loose or curly hair against a reasonably plain background (or `l03-curly-hair-portrait.jpg`). 16-bit, **Save As** `cutout-working.psd`. Duplicate the layer so the original stays intact beneath, and hide the original.
- 2 **Rough in the mass. Select ► Subject.** Tidy obvious misses with Quick Selection (paint to add, **Alt**/**⌘** to subtract). Don't chase strands yet.

- 3 **Refine the hairline.** Select ► **Select and Mask** (Ct r l / ⌘ + A l t / ⌥ + R). View: On Layers or Overlay. Edge Detection: tick **Smart Radius**, raise **Radius** a few pixels; try **Refine Hair**. Refine Edge Brush *along* the hairline. Smooth and Feather low; Shift Edge slightly negative.
- 4 **Rescue gaps.** Output To: **New Layer with Layer Mask** (Decontaminate off). Build a channel mask of the hair (best channel → duplicate → Levels), Ct r l / ⌘-click it, invert if needed. Click the layer mask thumbnail; paint **white** with a soft, low-opacity brush along the hairline.
- 5 **Decontaminate & pressure-test.** With the rescued mask selected, reopen Select and Mask, tick **Decontaminate Colors**, raise Amount, output to **New Layer with Layer Mask**. Put a saturated color layer *behind* the cutout; fix any fringe; place your real background (`103-new-background.jpg`). Save.

◆ RECIPE CARD: HARD CUTOUT

- 1 Select ► Subject for the solid mass
- 2 Select ► Select and Mask → Smart Radius on, Refine Edge Brush along the hairline; low Smooth/Feather, Shift Edge slightly negative; output New Layer with Layer Mask
- 3 Rescue missed strands: load a channel mask as a selection, paint white into the layer mask
- 4 Final pass: reopen Select and Mask on the rescued mask, tick Decontaminate Colors, output to New Layer with Layer Mask
- 5 Test against a saturated background; fix any fringe

Mantra: rough in, refine, rescue, decontaminate, in that order.

◆ YOUR TURN 1: SCORE YOUR EDGE AGAINST THE STANDARD

THE STANDARD	ON THE SATURATED TEST COLOR	ON THE NEW BACKGROUND
Fine strands kept		
No colored halo		
Soft (not razor) edge		

Which pipeline step fixed the most? Which strands did only the channel rescue bring back?

◆ YOUR TURN 2: HARDER CUTOUTS (HOMEWORK)

(a) A subject against a busy background (or `103-hair-busy-background.jpg`): where did the channel rescue earn its place, and where did you have to paint by hand? (b) Something non-human and tricky: a glass, a potted fern, a pet. Which part of the pipeline did it stress? (c) Put one cutout on a dark background and one on a light one. Where did a halo appear?

Going further: output to a Layer Mask, then later select that mask and reopen Select and Mask on it. What does that let you do?

◆ WATCH OUT

Hide the original layer before judging the edge: if it's visible underneath, every cutout looks perfect. Refine Edge Brush on the **boundary** only. Leave Decontaminate off until the final pass. The test color layer goes **below** the cutout. A layer "vanished"? Decontaminate hid it; check its eyeball.

◆ SELF-CHECK

Q1

What's the order of the hard-cutout pipeline?

A) Decontaminate, refine, rough in, rescue

B) Rescue, rough in, decontaminate, refine

C) Rough in, refine, rescue, decontaminate

D) Refine, decontaminate, rough in, rescue

Q2

Where does the Refine Edge Brush belong?

- A) Along the boundary between hair and background, never across the solid hair
- B) Over the whole subject, to smooth it
- C) Only on the face, to soften skin
- D) Across the solid interior of the hair, to add volume

Q3

In Select & Mask, what do Smart Radius and a slightly higher Radius do for hair?

- A) They sharpen the edge to a hard line
- B) They make Photoshop search a wider band around the edge for strands
- C) They remove the old background's color from the edge
- D) They feather the whole selection by that many pixels

Q4

Why is Decontaminate Colors left off for the first output?

- A) It only works on 8-bit files
- B) It deletes the flyaway strands
- C) It can only be used once per document
- D) It comes last: it rewrites edge pixels and forces a new layer, so you run it on the final, rescued mask

Q5

During the rescue, you've loaded the hair channel as a selection. What do you do next?

- A) Press Delete to remove the background inside the selection
- B) Click the layer mask thumbnail and paint white with a soft, low-opacity brush along the hairline
- C) Run Image ► Adjustments ► Levels on the RGB composite
- D) Click the image thumbnail and paint black over the missing strands

Q6

Your cutout looks fine on white but shows a halo on a dark background. What's the fix?

- A) Raise the Decontaminate Colors Amount, consider a tiny inward Shift Edge, and test on the darkest background it will sit on
- B) Raise Smooth to 100
- C) Flatten the image and export it on white
- D) Use Select Subject again and accept the result

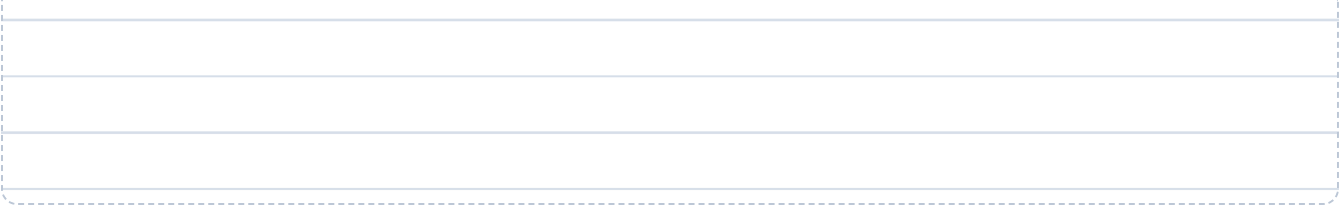
◆ JOURNAL

Which pipeline step did the heavy lifting on your image, and which did you almost skip but shouldn't have? Noticing that tells you where your cutouts usually go wrong. Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

◆ BEFORE THE NEXT MODULE

- ◆ Do three hard cutouts of increasing difficulty (studio hair, busy-background hair, a glass).
- ◆ Save a reusable Action for your Select and Mask output settings.
- ◆ Write your Learning Journal entry.

◆ NOTES



YOU FINISHED MODULE 1

Look at the arc: luminosity masks that let the image paint its own stencil, channel masks and Blend-If for subjects that tone or color makes distinct, and a full hard-cutout pipeline for flowing hair. Between them, almost nothing is truly unmaskable anymore. Cutting out hair used to be a rite of passage that took hours; you now have a repeatable four-step pipeline, and it carries into every composite you'll build. Next, Module 2 (Color & Tone, Deep): color management, grading with Curves, gradient maps and LUTs, and soft proofing for print.

Color & Tone, Deep

Build a color pipeline you can trust, author a cinematic grade and bottle it as a LUT, then split-tone, match, soft-proof and ship. Deliverable: a cinematic grade plus a print-ready, soft-proofed export: one look, delivered correctly to screen and to paper.

Lessons 2.1-2.3 (L04-L06)

3 lessons

Session 2 • 3 hours

Name: _____

HOW TO USE THIS WORKBOOK

Each lesson has **Key Ideas**, a **Worked Example** (the steps you'll follow in class), the lesson's **Recipe Card**, **Your Turn** exercises, a **Self-Check**, a **Journal** prompt, and space for **Notes**. Shortcuts are written Windows first, Mac second: **Ctrl**/**⌘** means Ctrl on Windows, Command on a Mac.

TWO RULES FOR THIS WHOLE MODULE

1. **Judge color on your own screen**, not the projector. Turn off Night Light / True Tone / Night Shift and set brightness to about the middle. 2. **When Photoshop asks about a profile, read the dialog before you click**. On open: use the embedded profile. At the end: **Convert** (never Assign), on a duplicate.

MODULE PROGRESS TRACKER

- ◆ ☐ **L04** I set up Color Settings, saw 8-bit banding vs 16-bit smoothness, and exported a web copy Converted to sRGB
- ◆ ☐ **L05** I built a grade on a neutral base, saved `grade-working.psd`, and exported a LUT that worked on a second photo
- ◆ ☐ **L06** I split-toned, matched a companion image, soft-proofed, and delivered a web file and a print file (the Module 2 deliverable)
- ◆ ☐ I wrote a journal entry after each lesson

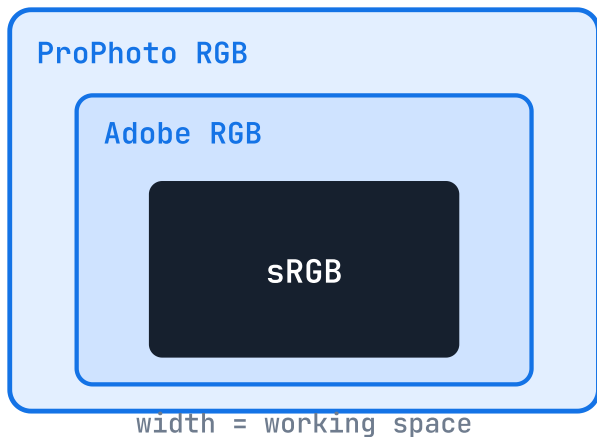
◆ BY THE END YOU CAN

- ◆ Explain working spaces (sRGB, Adobe RGB, ProPhoto) and pick the right one
- ◆ Say why you edit in **16-bit** and what **banding** actually is
- ◆ Read the histogram and **diagnose an image before touching a slider**
- ◆ Set up Photoshop's **Color Settings** once, correctly, for the rest of the course

◆ KEY IDEAS

- ◆ **An image is color in a container** with two properties: how **wide** it is (the working space, or gamut) and how **finely divided** (the bit depth). 8-bit = 256 steps per channel; 16-bit = over 32,000.
- ◆ **Banding** is what happens when an edit demands more tonal steps than the file has. 16-bit has about 128 times more steps, so gradients survive hard grading.
- ◆ A number like "R 220" is meaningless without its container. Color management keeps track of the container so a color means the same thing from capture to screen to print.
- ◆ **Capture wide, edit wide and deep, convert down last.** Never edit in the narrow export space.
- ◆ **Convert to Profile** remaps the colors to fit the new space and keeps them looking right. **Assign Profile** just relabels the container and shifts your colors.

SPACE	USE FOR	NOTES
sRGB	Web, social, most screens, email	The safe default for anything online. Assume the internet is sRGB.
Adobe RGB	Print work, especially CMYK	Holds the cyans and greens many printers can reproduce but sRGB clips.
ProPhoto RGB	Heavy RAW grading, archival masters	Enormous gamut. Always pair with 16-bit or you will band.



16-bit: smooth after a hard edit



8-bit: the same edit bands

Width is the working space; the fineness of the steps is the bit depth.

◆ WORKED EXAMPLE: SET UP & DIAGNOSE

- 1 **Edit ► Color Settings.** RGB working space: **Adobe RGB (1998)** for print-capable work (or sRGB if you are strictly web). CMYK: your print provider's press profile (for example Coated GRACoL 2006 or Coated FOGRA39) or the default. Policies: **Preserve Embedded Profiles**, mismatch warnings on (**Ask When Opening**).
- 2 Open your image (or `104-smooth-sky-16bit.tif`). Confirm the bit depth: **Image ► Mode ► 16 Bits/Channel** for anything you will grade hard.
- 3 **Window ► Histogram.** Before any edit: crammed left (underexposed) or right (overexposed)? A spike against either wall (clipped, lost detail)? Narrow (flat, low contrast)? Write one sentence: what does the image need?
- 4 **Prove the container matters.** Duplicate a smooth-gradient image; set one copy to 8-bit, keep the other 16-bit. Apply the same aggressive Curves to both, then a strong brighten. Zoom in: the 8-bit copy bands.
- 5 **Export correctly.** Save the layered master (for example `color-master.psd`). For web: **Image ► Duplicate** (tick **Duplicate Merged Layers Only**), then on the copy **Edit ► Convert to Profile ► sRGB**, **Image ► Mode ► 8 Bits/Channel**, export. (**File ► Export ► Export As** can also convert to sRGB for you.)

◆ RECIPE CARD: TRUSTWORTHY COLOR, START TO FINISH

- 1 **Edit ► Color Settings:** deliberate working space, Preserve Embedded Profiles
- 2 Work in 16-bit for anything you will grade hard
- 3 Read the histogram and name the problem before editing
- 4 Keep the master wide + 16-bit
- 5 **Export:** on a duplicate, Convert to Profile (sRGB for web) at the very end

Mantra: capture wide, edit deep, convert down last.

◆ YOUR TURN 1: MY COLOR SETTINGS

Write down what you chose, and why, so you can set up any machine the same way.

RGB working space

CMYK

RGB policy

Mismatch / missing boxes ticked

Why this working space for my work

◆ YOUR TURN 2: DIAGNOSE BEFORE YOU EDIT

Read each histogram **cold** (`104-histogram-a` to `e`), or five of your own images). Guess from the histogram alone, then look at the image and check. Careful: one of them is fine as it is.

FILE	WHAT THE HISTOGRAM SHOWS	MY ONE-SENTENCE EDIT PLAN
a		
b		
c		
d		
e		

My own image's diagnosis, in one sentence:

◆ YOUR TURN 3: SEE THE SHIFT (NOW YOU, FINISH AT HOME)

Start from an Adobe RGB (or ProPhoto) version, such as `104-lagoon-adobergb.jpg` , and export it two ways: correctly Converted to sRGB, and deliberately with the profile stripped. View both in a browser. (Stripping an sRGB file shows no shift, because browsers assume sRGB.) Describe the shift, and where you saw it most:

Then set up a wide-gamut, 16-bit document template you can start pro work from. Its settings:

◆ WATCH OUT

Assign just relabels and will shift your colors; **Convert** keeps the appearance. Do the conversion on a **duplicate**, so your master stays wide and 16-bit. ProPhoto in 8-bit bands badly.

◆ SELF-CHECK

Q1

An image is "color poured into a container." Which two properties define that container?

- A) File format and pixel dimensions
- B) Working space (how wide the gamut is) and bit depth (how finely each channel is divided)
- C) Color mode and the number of layers
- D) Monitor brightness and contrast

Q2

You apply the same aggressive Curves to an 8-bit and a 16-bit copy of a smooth sky. Only the 8-bit copy breaks into bands. Why?

- A) The 16-bit copy has more pixels
- B) 8-bit files always use a smaller working space
- C) Photoshop turns off Curves smoothing in 8-bit
- D) The edit demands more tonal steps than 8-bit's 256 per channel; 16-bit has about 128 times more

Q3

A learner's graded photo looks duller on Instagram than in Photoshop. What is the most likely cause?

- A) The file was left in Adobe RGB or ProPhoto and never Converted to sRGB
- B) The JPEG quality was set too low
- C) The master was saved in 16-bit
- D) Preserve Embedded Profiles was turned on

Q4

It is time to deliver a web JPEG from your layered, 16-bit Adobe RGB master. Which is the correct move?

- A) Edit ► Assign Profile ► sRGB on the master, then save
- B) Image ► Mode ► 8 Bits/Channel on the master, then export
- C) On a merged duplicate: Edit ► Convert to Profile ► sRGB, 8 Bits/Channel, then export
- D) Save the layered master directly as a JPEG

Q5

Before any edit, the Histogram shows a tall spike slammed against the right wall. What does it tell you?

- A) The image is underexposed
- B) Highlights are clipped: detail at the bright end is lost
- C) The image is flat and low in contrast
- D) It is a correctly exposed high-key image with nothing to fix

Q6

Which pairing of working space and job matches the lesson's table?

A) ProPhoto RGB for web graphics, in 8-bit

B) sRGB for CMYK print work

C) Adobe RGB for email newsletters

D) ProPhoto RGB for heavy RAW grading and archival masters, always paired with 16-bit

◆ JOURNAL

Have you ever had colors "change" on export or a print come back wrong? Now that you understand containers, what was probably going on? Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

IF THIS LESSON FELT DRY

That's normal: most people glaze over at color management. Pushing through it is exactly what separates hobby edits from professional ones. Everything downstream in this module now lands the way you intend.

◆ NOTES

◆ BY THE END YOU CAN

- ◆ Grade with **per-channel Curves** for precise color in shadows, mids and highlights
- ◆ Use a **Gradient Map** to remap tones to a color ramp
- ◆ Apply and save a **LUT** (Color Lookup) to reuse a grade consistently
- ◆ Keep the whole grade non-destructive and dialable

◆ KEY IDEAS

- ◆ Correcting asks "what is wrong?" **Grading asks "what should it feel like?"** Every grade is a remap: you decide what color each tone becomes.
- ◆ Three remapping tools, from surgical to sweeping: **per-channel Curves** (the precision instrument), **Gradient Map** (map the whole tonal range to a ramp), **LUT** (a saved recipe of a finished grade).
- ◆ **Gradient Map**: darkest pixels get the gradient's left color, brightest get its right color. **Blend it, don't bake it**: Color or Soft Light, opacity around 15–40%.
- ◆ **LUT** (Look-Up Table, a `.cube` or `.3dl` file): "turn this input color into that output color," for every color. Apply one with a **Color Lookup** adjustment layer.
- ◆ **Correct, then grade, then bottle.** Grading on top of a neutral base is what makes a look portable.



Correct, then grade, then bottle.

◆ WORKED EXAMPLE: GRADE & BOTTLE IT

- 1 Open a portrait (or `l05-portrait-studio.jpg`) in your 2.1 pipeline: 16-bit, deliberate working space. Correct to a **neutral base** first (exposure and white balance) with a Curves layer or **Filter ▶ Camera Raw Filter**. Save As `grade-working.psd`.
- 2 **Contrast, then color.** Curves layer: a gentle RGB S-curve. New Curves layer, **Blue** channel: lift the shadow end (blue shadows), drop the highlight end (warm, yellow highlights). For a truer teal, also lift Green slightly or drop Red slightly in the shadows.

- 3 **Gradient Map accent.** Teal-to-gold ramp, layer set to **Color** or **Soft Light**, opacity about 20–30%.
- 4 **Refine locally.** Mask a Curves or the Gradient Map layer so skin stays natural while the background goes moody. A luminosity mask (Lesson 1.1) keeps the grade out of the brightest highlights.
- 5 **Bottle it.** Save `grade-working.psd` first. Disable the masks (Shift-click each mask thumbnail), hide anything that is not part of the grade, keep a Background layer plus adjustment layers, then **File ► Export ► Color Lookup Tables** → save a `.cube`. Re-enable the masks.
- 6 Open a different photo (or `105-pier-dusk.jpg`), add a **Color Lookup** layer, **Load 3D LUT...**, choose your `.cube`. Watch the whole look transfer.

◆ RECIPE CARD: BUILD A LOOK AND SAVE IT

- 1 Curves: contrast (RGB S-curve) first
- 2 Curves per channel: teal shadows, warm highlights (Blue channel split)
- 3 Gradient Map (teal→gold) at ~20–30% / Color blend
- 4 Mask locally (luminosity or subject masks)
- 5 Save the master; disable masks, then **File ► Export ► Color Lookup Tables** → reuse as a `.cube`

Mantra: correct, then grade, then bottle.

◆ YOUR TURN 1: SKETCH YOUR BLUE CURVE

Draw the Blue-channel curve you ended up with (mark the shadow end and highlight end), and note any Green or Red moves you added for a truer teal:

◆ YOUR TURN 2: THREE LUTS (NOW YOU, START NOW, FINISH AT HOME)

Build three distinct LUTs and apply each to the same photo. Fill in the recipe for each:

LOOK	SHADOW COLOR	HIGHLIGHT COLOR	CONTRAST	.CUBE FILE NAME
Warm summer				
Cold thriller				
Faded matte				

Apply one LUT across five different photos. Where did it need per-image tweaking, and why?

◆ YOUR TURN 3: REVERSE-ENGINEER A LOOK

Pick a film still or a favorite photo. Name its shadow color and highlight color, then recreate it with per-channel Curves. Which channels did you bend, and which way?

◆ WATCH OUT

A Gradient Map at 100% Normal looks garish and kills detail. The LUT export bakes **every visible** adjustment layer and cannot store a mask, so disable masks and hide non-grade layers first, and remember to re-enable the masks and save afterward.

◆ SELF-CHECK

Q1

What separates grading from correcting, in the lesson's words?

- A) Grading only happens in Camera Raw
- B) Grading changes exposure; correcting changes color
- C) Correcting asks "what is wrong?"; grading asks "what should it feel like?" and deliberately remaps tones to chosen colors
- D) There is no difference; they are two names for one step

Q2

You want the classic teal-and-gold in Curves. In the Blue channel, what do you do?

- A) Lift the shadow end and drop the highlight end
- B) Drop the shadow end and lift the highlight end
- C) Raise only the midpoint
- D) Flatten the curve to a horizontal line

Q3

Your teal-to-gold Gradient Map looks garish and has flattened all the detail. What does the lesson tell you to do?

- A) Delete it; Gradient Maps are only for black-and-white work
- B) Rasterize it so it merges with the photo
- C) Swap the gradient's end colors
- D) Set the layer to Color or Soft Light and/or drop its opacity to about 15–40%

Q4

Before File ► Export ► Color Lookup Tables, why do you Shift-click each mask thumbnail and hide non-grade layers?

- A) The export only works on flattened images
- B) The export bakes every visible adjustment layer, and a LUT cannot store a mask
- C) Masks make the .cube file too large to load
- D) Color Lookup Tables only export the selected layer

Q5

Your LUT looks perfect on the shot you built it on but wrong on another photo. What is the most likely reason?

- A) The second photo wasn't corrected to a neutral base first, and a LUT assumes a similar starting point
- B) LUTs only work on the image they were built on
- C) The .cube file was saved in the wrong folder
- D) The Color Lookup layer needs to be at 100% opacity

Q6

Which order does the lesson's flowchart and build follow?

A) Bottle the LUT → correct → grade

B) Color → contrast → correct → mask

C) Correct to neutral → contrast (S-curve) → color → refine locally with masks → bottle as a LUT

D) Mask → Gradient Map → correct → contrast

◆ JOURNAL

Describe the "look" you are drawn to as an editor in terms of shadow color, highlight color, and contrast.

Naming your taste is the first step to building it into a LUT. Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

◆ NOTES

◆ BY THE END YOU CAN

- ◆ **Match color** across two images so a set feels like one shoot
- ◆ Refine a **split-tone** (warm highlights, cool shadows) with Color Balance / Curves
- ◆ **Soft-proof** for a printer and read the gamut warning
- ◆ Deliver a graded image as both a **web file** and a **print-ready file**

◆ KEY IDEAS

- ◆ A **split-tone** puts one color into the shadows and a different (usually complementary) color into the highlights. The classic: **cool shadows**, **warm highlights**, mids roughly neutral so skin stays believable.
- ◆ **Color Balance** has explicit Shadows / Midtones / Highlights buttons; per-channel Curves do the same by which end of the curve you bend.
- ◆ **Matching**: sample a known neutral (or skin) in each image with the **Color Sampler Tool**, read the **Info panel** (Window ► Info), and move each Curves channel until the values meet. **Match tones before colors**: black point, white point, then mid color.
- ◆ **Soft-proofing** simulates the print on screen: **View ► Proof Setup ► Custom**. Toggle proof with **Ctrl/⌘ + Y**; **Gamut Warning** (**Ctrl/⌘ + Shift + Y**) grays out colors the printer can't reproduce.
- ◆ **Neon dies on paper**. Pull saturation down where the warning lights up, and the print matches the proof.



cool shadows

neutral mids (skin)

warm highlights

one color into the dark end, its complement into the bright end

Split-tone by zone: let the extremes carry the color story.

CUSTOMIZE PROOF CONDITION	SETTING
Device to Simulate	Your printer's profile (stand-in: U.S. Web Coated (SWOP) v2)
Rendering Intent	Relative Colorimetric or Perceptual
Black Point Compensation	On

Simulate Paper Color

On (the image dulls: that's the paper)

◆ WORKED EXAMPLE: SPLIT-TONE & PROOF

- 1 Open `grade-working.psd` from Lesson 2.2 (16-bit, wide space). If you graded a landscape there, drag your Lesson 2.2 grading layers onto a corrected portrait instead. **Save As** `finish-working.psd`.
- 2 **Split-tone.** Add a **Color Balance** layer. **Shadows:** push toward cyan/blue a little. **Highlights:** toward red/yellow. **Midtones:** near neutral. Judge on skin: warmly lit, not sunburnt.
- 3 **Match a companion.** Open a second photo of the same set (or `106-companion-portrait.jpg`). Drop a Color Sampler on a comparable point in each. Add Curves to the second image and move each channel until its sampled values approach the first image's.
- 4 **Soft-proof.** **View ▶ Proof Setup ▶ Custom:** printer profile (or SWOP v2), Relative Colorimetric or Perceptual, Black Point Compensation, Simulate Paper Color. Proof on (`Ctrl/⌘ + Y`). Gamut Warning on (`Ctrl/⌘ + Shift + Y`); lower the saturation of anything gray with a masked Hue/Saturation until it clears.
- 5 **Deliver both files.** Save the layered master first. **Web:** Image ▶ Duplicate (Duplicate Merged Layers Only), Convert to Profile ▶ sRGB, 8 Bits/Channel, export a JPG. **Print:** another merged duplicate, Convert to Profile ▶ your printer/CMYK profile (or keep tagged RGB if your printer asks for it), save as PDF or TIFF.

◆ RECIPE CARD: SPLIT-TONE, MATCH, AND PRINT-CHECK

- 1 Color Balance: cool shadows, warm highlights, neutral mids
- 2 Match a companion image with Color Sampler + Curves (tones before colors)
- 3 View ▶ Proof Setup ▶ Custom → printer profile, Rendering Intent + Black Point Compensation, Simulate Paper Color
- 4 Gamut Warning (`Ctrl/⌘ + Shift + Y`) → desaturate flagged areas
- 5 Deliver from duplicates: Convert to sRGB (web) and to the printer profile (print)

Mantra: what you see is only trustworthy once you have proofed it.

◆ YOUR TURN 1: READ THE SAMPLERS

Record the Info panel readouts for the same point (gray card middle patch or cheek) before and after your Curves match.

POINT

REFERENCE R / G / B

COMPANION BEFORE

COMPANION AFTER

#1 _____

#2 _____

Which channel had to move the most, and which way?

◆ YOUR TURN 2: DELIVERABLE LOG

Master file / mode

finish-working.psd · RGB/16 · _____

Proof profile + intent

What the gamut warning flagged

Web file

Print file (profile, format)

◆ YOUR TURN 3: NOW YOU (HOMEWORK)

Match a set of three photos shot in different light (106-set-a-daylight , -b-tungsten , -c-shade , or your own) so they could sit together on a page. Then split-tone one image two ways: cinematic cool/warm, and an unusual pairing (green shadows, magenta highlights). How did the pairing change the emotion?

Soft-proof a deliberately vivid image (or 106-neon-proof.jpg). How much saturation did you have to surrender for print?

◆ WATCH OUT

Color Balance opens on **Midtones**: switch to Shadows and Highlights before you push. If skin looks ill, the color is reaching the mids: ease off, and mask the grade off the face. Never convert the master to CMYK; always a merged duplicate.

◆ SELF-CHECK

Q1

What is the most flattering, cinematic split-tone according to the lesson?

- A) Warm shadows, cool highlights, saturated mids
- B) Cool shadows, warm highlights, with the mids kept roughly neutral
- C) The same color in shadows and highlights
- D) Green shadows and magenta highlights

Q2

After split-toning, the skin looks ill. What is going on and what is the fix?

- A) The image needs converting to CMYK
- B) The Gradient Map from Lesson 2.2 must be deleted
- C) Preserve Luminosity is ticked; untick it
- D) The color is reaching the midtones where skin lives: keep the split in the extremes, ease the amount, and mask the grade off the face if needed

Q3

You are matching a companion photo to a reference with Curves. What order does the lesson give?

- A) Tones first (brightness and contrast), then black point, white point and mid color from the sampled values
- B) Mid color first, then tones
- C) Saturation first, then white balance
- D) Run Match Color, then stop

Q4

Where do you read the RGB values of the points you dropped with the Color Sampler Tool?

- A) The Histogram panel
- B) The Properties panel of the Curves layer
- C) The Info panel (Window ► Info)
- D) The Layers panel

Q5

With soft-proofing on, Gamut Warning (Ctrl/⌘ + Shift + Y) grays out a neon sign. What do you do?

- A) Nothing; the printer will fix it
- B) Lower the saturation there with a masked Hue/Saturation layer until the warning clears
- C) Assign the printer profile to the master
- D) Turn off Simulate Paper Color

Q6

Which delivery matches the Module 2 deliverable?

- A) One sRGB JPEG, with the master flattened to save space
- B) Assign CMYK to the master and export a JPEG
- C) Convert the layered master to 8-bit sRGB and save over it
- D) Keep the layered master wide + 16-bit; from merged duplicates, Convert to sRGB (8-bit JPG) for web and to the printer's profile (PDF or TIFF, or tagged RGB if the printer asks) for print

◆ JOURNAL

If you have ever printed a photo that disappointed you, what do you now suspect went out of gamut? How would soft-proofing have caught it? Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

◆ BEFORE THE NEXT MODULE

- ◆ Deliver one image as both a web and a print-ready file.
- ◆ Match a small set of photos into one consistent look.
- ◆ Write your Learning Journal entry.
- ◆ Bring a high-resolution portrait you have permission to retouch (Module 3).

◆ NOTES

YOU FINISHED MODULE 2

You built a trustworthy color pipeline, authored and bottled a cinematic grade, and finished with split-toning, matching and print-safe soft-proofing. You can now take color from a raw capture to a graded look that holds up on any screen and on paper. That end-to-end command of color is rare, and highly employable. In Module 3 (High-End Retouching) we turn to people: frequency separation, dodge and burn, and a natural finish.

High-End Retouching

The professional portrait pipeline, on one face: even the skin with frequency separation while keeping every pore, sculpt the face back into three dimensions with dodge and burn, and finish the eyes, teeth, and hair so gently that a stranger calls it "a great photo," not "a great retouch." The deliverable is a magazine-quality portrait that still looks exactly like the person.

Lessons L07–L09 (3.1–3.3)

3 lessons

Session 3 • 3 hrs

Name: _____

HOW TO USE THIS WORKBOOK

Each lesson has goals, **Key Ideas**, the **Guided Build** steps, **Your Turn** exercises, a six-question **Self-Check**, the lesson's **Journal** prompt, and **Notes**. Work on a self-portrait, a portrait of someone who agreed to be edited *and* shown in class, or the practice portraits ([labs/module-3/practice/](#)). When we critique, we talk about the *edit*, never about anyone's looks.

MODULE PROGRESS TRACKER

- ◆ ☐ **L07 (3.1)**: I built a correct Low/High separation for my file's bit depth, evened the tone, healed the blemishes, and saved `skin-retouched.psd` with real pores at 100%
- ◆ ☐ **L08 (3.2)**: I sculpted the face on a 50% gray Soft Light layer, checked it in black and white, and saved `portrait-sculpted.psd`
- ◆ ☐ **L09 (3.3)**: I finished the eyes, teeth, and hair through masks and passed the flip-and-mirror test (Module 3 deliverable)

◆ BY THE END YOU CAN

- ◆ Explain what frequency separation splits an image into and why it matters
- ◆ Build a correct low-frequency (tone/color) and high-frequency (texture) pair
- ◆ Even out blotchy skin on the low layer without smearing pores
- ◆ Remove a blemish on the high layer without leaving a soft patch
- ◆ Know when frequency separation is the wrong tool

◆ KEY IDEAS

- ◆ Skin is two things layered together. **Low frequency**: slow, gradual changes (cheek to shadow, redness, an uneven tan): **tone and color**. **High frequency**: fast, tiny changes (pores, fine hairs, grain): **texture**.
- ◆ A blur or a soft brush destroys both at once: melted plastic. Frequency separation puts them on two layers so each edit never spills into the other.
- ◆ **Low** = a copy blurred until the pores vanish but the features stay (often 4–8 px). **High** = Apply Image from Low, set to **Linear Light**. Together they look identical to the original.
- ◆ **Apply Image depends on bit depth**. 8-bit: Blending **Subtract**, Scale **2**, Offset **128**. 16-bit (the course's recommendation): Blending **Add**, **Invert** ticked, Scale **2**, Offset **0**. There is no second radius to enter; Apply Image subtracts Low itself.
- ◆ **The radius follows pore size in pixels**. A close beauty shot has bigger pores in the frame, so it needs a larger radius (and more restraint) than a candid at arm's length.
- ◆ Group the pair and lower the group to about **70–85%**. A little of the original showing through keeps it human.



High • texture only

[Linear Light](#) • [heal blemishes here](#)



Low • tone & color

Normal • Gaussian Blur • even tone here

Background • untouched safety copy 

Group Low + High and lower the group's opacity to taste. Each edit uses Sample: Current Layer.

◆ GUIDED BUILD: RETOUCH THE SKIN

- 1 Open your portrait and do your global corrections first (Module 2). Check the bit depth in the title bar or **Image ► Mode**.
- 2 **Make the layers (10 min)**. Duplicate the layer twice (**Ctrl/⌘ + J**). Lower copy = **Low**, upper = **High**. Hide High. On Low: **Filter ► Blur ► Gaussian Blur** until the pores vanish but the features stay. Note the radius.
- 3 Show and select High: **Image ► Apply Image**, Layer **Low**, Channel **RGB**. 16-bit: **Add**, **Invert** ✓, Scale **2**, Offset **0**. 8-bit: **Subtract**, Scale **2**, Offset **128**. Blend mode → **Linear Light**. ✓ Checkpoint: it looks normal again.
- 4 **Even the tone on Low (12 min)**. Big, soft Healing Brush (**Sample: Current Layer**) or a low-opacity Mixer Brush (Sample All Layers off). Push blotchy color toward its neighbors: redness by the nose, a harsh tan line, a muddy shadow.
- 5 **Remove blemishes on High (10 min)**. Healing Brush or Clone Stamp, **Sample: Current Layer**. Sample clean texture near a blemish and heal over it.
- 6 **Dial it back (5 min)**. Group both layers (**Ctrl/⌘ + G**), lower the group to 70–85%. Zoom to 100%: real pores everywhere? Save as **skin-retouched.psd** with layers. Lesson 3.2 opens this file.

◆ YOUR TURN: WHICH LAYER FIXES IT?

Survey your portrait (or **107-portrait-retouch.jpg**) at 100% *before* touching a tool. Sort every problem you see into the layer that fixes it. Anything that is part of the person (freckles, a mole, laugh lines) goes in neither column: it stays.

Low: tone & color problems

High: texture problems

◆ YOUR TURN: THE RADIUS FOLLOWS THE PORES (NOW YOU)

On each file, raise the Gaussian Blur on Low until the pores just vanish. Record the radius and one sentence on how much restraint that face needs.

Main portrait (<code>107-portrait-retouch.jpg</code> or mine)	_____	_____
	px	
Close beauty shot (<code>107-beauty-closeup.jpg</code> or mine)	_____	_____
	px	
Candid at arm's length (<code>107-candid-distance.jpg</code> or mine)	_____	_____
	px	

At home: over-smooth one face on purpose, then rescue it with group opacity. Then try the same fix with a plain soft brush and no separation. What does the comparison tell you?

◆ WATCH OUT

Stack doesn't match the original? Check the bit depth: the 8-bit recipe on a 16-bit file fails. **High shows eyes and lips?** The radius was too large; rebuild smaller. **High almost blank?** Too small; the pores stayed on Low. **Color smear after a heal on High?** The Sample setting wasn't Current Layer.

◆ SELF-CHECK

Q1

In frequency separation, what does the low-frequency layer hold?

- A) Pores, fine hairs, and the grain of the skin
- B) Tone and color: the gradual shifts, patches of redness, and shading
- C) Only the edges of the eyes, nose, and lips
- D) A neutral gray map of the texture

Q2

How far should you raise the Gaussian Blur on the Low layer?

- A) Until the eyes and lips are soft too, so nothing sharp is left
- B) Always 20 px, whatever the photo
- C) Until the whole layer turns flat gray
- D) Just until the pores vanish but the features stay, and note the radius

Q3

Your file is 16 Bits/Channel. Which Apply Image settings build the High layer from Low?

- A) Blending Add, Invert ticked, Scale 2, Offset 0
- B) Blending Subtract, Scale 2, Offset 128
- C) Blending Add, Scale 1, Offset 128
- D) Blending Subtract, Invert ticked, Scale 1, Offset 0

Q4

You want to calm the redness around the nose without moving any pores. Where and how?

- A) On High, with the Healing Brush set to Sample: All Layers
- B) On the Background, with a soft Brush at 100% opacity
- C) On Low, with a big soft Healing Brush (Sample: Current Layer) or a low-opacity Mixer Brush
- D) On High, with a small Gaussian Blur over the nose

Q5

Your High layer shows the eyes, lips, and the shape of the face, not just fine texture. What happened?

- A) The blur radius on Low was too large, so real features moved into the texture layer
- B) The blur radius on Low was too small
- C) High's blend mode is Linear Light
- D) The file is 8-bit

Q6

You retouch a close beauty shot and a candid taken at arm's length. Which usually needs the larger blur radius?

- A) The candid, because the face is smaller
- B) Neither: the radius is always the same for skin
- C) Whichever file has the higher bit depth
- D) The close beauty shot, because its pores are bigger in pixels

◆ JOURNAL

Look at a magazine cover and a phone-beauty-filter selfie. Both smooth skin; describe, in frequency terms, why one reads as expensive and the other as fake. Also note: key concepts · what clicked · questions · ideas to try · how you feel about your progress.

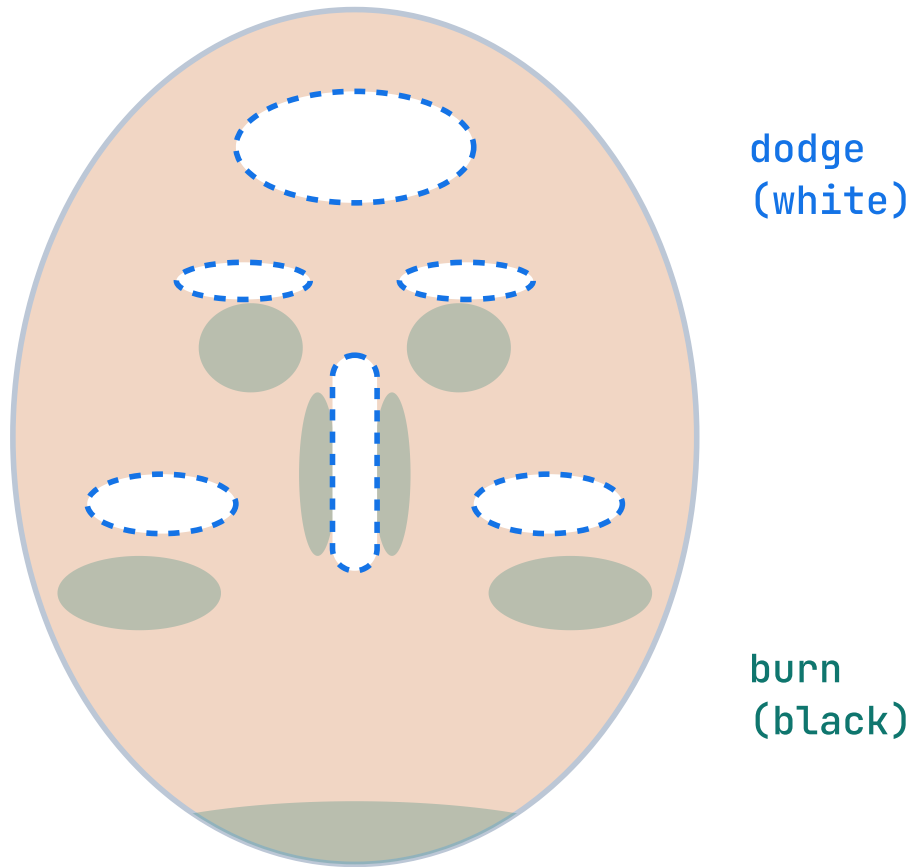
◆ NOTES

◆ BY THE END YOU CAN

- ◆ Explain why dodge and burn adds form that smoothing can't
- ◆ Build a non-destructive 50% gray Soft Light dodge and burn layer
- ◆ Work at low flow to build light gradually instead of streaking it on
- ◆ Sculpt the planes of a face: cheekbones, jaw, nose bridge, brow
- ◆ Check your work in black and white to judge form without color

◆ KEY IDEAS

- ◆ A camera flattens a face; our brains read depth from where light falls. **Dodging** lightens and pulls a plane toward the viewer. **Burning** darkens and pushes it back.
- ◆ **Low and slow.** Brush over a plane many times at Flow 5–10%. Streaky dodge and burn is just impatient dodge and burn.
- ◆ Paint on a **50% gray layer in Soft Light**: gray is invisible, lighter than gray dodges, darker burns. Every stroke stays adjustable, reversible, and maskable. The Dodge and Burn tools edit pixels directly and are hard to undo.
- ◆ **Follow the light that's already there.** You amplify it; you don't invent a new light source.
- ◆ Color hides uneven sculpting: check form under a temporary **Black & White** adjustment layer. Finish by lowering the gray layer's opacity until the effect is **felt, not seen**.



Dashed white = lift what should catch light. Shaded = deepen what should recede. Follow the bones.

◆ GUIDED BUILD: SCULPT THE FACE

- 1 Open `skin-retouched.psd` from L07 and select the top of the stack so the new layer lands above the retouch (above the L07 group, not inside it).
- 2 **Add the gray layer (3 min).** Layer ► New ► Layer, Mode **Soft Light**, tick **Fill with Soft-Light-neutral color (50% gray)**. Soft round brush, **Hardness 0**, **Flow 5–10%**, **Opacity 100%**. White = dodge, black = burn (X swaps).
- 3 **Burn the recesses (12 min).** Black: eye sockets, sides of the nose, under the cheekbones, under the jaw. Many light passes.
- 4 **Dodge the highlights (12 min).** White: tops of the cheekbones, bridge and tip of the nose, center of the forehead, brow bone. Amplify the existing light.
- 5 **Judge and temper (6 min).** Toggle a temporary Black & White adjustment layer on top, fix lumps and streaks, turn it off. Lower the gray layer's opacity until the sculpting is felt, not seen. Delete the B&W layer and save as `portrait-sculpted.psd`. Lesson 3.3 finishes this file.

Want to see only your light? **Alt/⌘**-click the gray layer's eye to show it alone, then again to return.

◆ YOUR TURN: READ THE LIGHT FIRST

Before you paint: where does the light in your portrait come from? Which side of the face is brighter? Then plan your map.

Light comes from...	_____
Three planes I will dodge (most important first)	_____
Three planes I will burn	_____
What the Black & White check showed me	_____
Final gray-layer opacity	_____ %

◆ YOUR TURN: SAME FACE, THREE MOODS (NOW YOU)

Sculpt one flatly lit portrait three ways, light only. Note where the dodge and burn moved for each mood. (Or sculpt `108-apple-flat.jpg`, a bag, or a car panel instead: same principle.)

Soft beauty / gentle top light

Dramatic side light

◆ WATCH OUT

Solid gray over the photo? The mode is Normal; set Soft Light. **Streaks?** Flow too high or the brush is hard. **Paint on the face?** The photo layer was selected. **Everyone can see where you dodged?** It's too strong: lower the layer's opacity.

◆ SELF-CHECK

Q1

On a 50% gray layer set to Soft Light, what does painting with black do?

- A) Lightens that area and pulls it toward the viewer
- B) Nothing, because black is invisible in Soft Light
- C) Darkens (burns) that area and pushes that plane back
- D) Paints solid black over the face

Q2

Which brush setup matches the lesson?

- A) Soft round, Hardness 0, Flow 5–10%, Opacity 100%
- B) Hard round, Hardness 100, Flow 100%, Opacity 100%
- C) Hard round, Flow 100%, Opacity 10%
- D) The Dodge tool at Exposure 100%

Q3

According to the lesson's face map, which planes do you burn?

- A) The tops of the cheekbones and the bridge of the nose
- B) The center of the forehead and the brow bone
- C) The tip of the nose and the catch-lights
- D) The eye sockets, the sides of the nose, under the cheekbones, and under the jaw

Q4

Why add a temporary Black & White adjustment layer while you work?

- A) The final portrait must be delivered in black and white
- B) Color hides uneven sculpting; without it, lumps and streaks in the light become obvious
- C) Soft Light only works on grayscale images
- D) It makes the 50% gray layer visible so you can paint on it

Q5

You finish sculpting and a friend can point to exactly where you dodged. What's the lesson's fix?

- A) Duplicate the gray layer to even it out
- B) Switch the gray layer to Normal mode
- C) Lower the gray layer's opacity until the sculpting is felt but not seen
- D) Paint more white on the opposite side to balance it

Q6

For client work you might revise, why does the gray layer beat the Dodge and Burn tools?

- A) It's non-destructive and maskable, and you can dial the whole effect up or down afterward
- B) The Dodge and Burn tools only work on 16-bit files
- C) The gray layer is faster because it needs only one stroke
- D) The tools can't lighten, only darken

◆ JOURNAL

Study a portrait you admire and, without touching the file, describe where its retoucher likely dodged and where they burned. Training your eye to see the light is half the skill. Also note: key concepts · what clicked · questions · ideas to try · how you feel about your progress.

◆ NOTES

◆ BY THE END YOU CAN

- ◆ Brighten and sharpen eyes with a masked local edit
- ◆ Whiten teeth by removing yellow first, then brightening, not the reverse
- ◆ Tame flyaway hair and fill gaps without a helmet-hair look
- ◆ Apply every finish through a mask so nothing bleeds where it shouldn't
- ◆ Judge the "natural line": the point where more becomes worse

◆ KEY IDEAS

- ◆ **The last five percent** (brighter eyes, cleaner teeth, tamed hair) is what people actually notice. Restrained, it lifts a portrait; overdone, it screams "edited."
- ◆ Every finish has one shape: **make an adjustment, hide it behind a black mask, paint it back only where it belongs.** Nothing global, everything local.
- ◆ **Teeth: yellow out first, brightness second.** Brightening a yellow tooth gives you a bright yellow tooth.
- ◆ Real eye whites are faintly gray or pink; real teeth are faintly warm. Aim for **clean, never glowing.** The catchlight is sacred.
- ◆ **Hair:** clone flyaways out and fill gaps on their own layer, following the hair's flow, and leave a few strays. Perfectly smooth hair looks like a wig.
- ◆ The pipeline order is the workflow: **global corrections → frequency separation → dodge & burn → eyes/teeth/hair.**

**Eyes • Curves (lift) + High Pass**

black mask • paint iris & whites

**Teeth • Hue/Sat (-yellow, +light)**

black mask • paint teeth only

**Hair • empty layer (Clone / Heal)**

Sample: All Layers • flyaways out, gaps in

Three small, separate edits. Any one can be dialed back or deleted without disturbing the others.

◆ GUIDED BUILD: FINISH THE PORTRAIT

- 1 Open **portrait-sculpted.psd** from L08.
- 2 **Eyes (12 min)**. Curves layer, lift the midtones a little. Invert the mask (Ctrl/⌘ + I), paint white over the iris and whites with a small soft brush. Optional sharpen: merged copy (Ctrl + Alt + Shift + E / ⌘ + ⌥ + ⬆ + E), **Filter ▶ Other ▶ High Pass** about 1–3 px, blend **Overlay** or **Soft Light**; Alt/⌥-click Add Layer Mask for a black mask, paint white over the iris only. Never paint out the catchlight.
- 3 **Teeth (10 min)**. Hue/Saturation layer: **Yellows**, drop the saturation; then **Master**, raise the lightness slightly. Invert the mask, paint the teeth only, feather the gum line.
- 4 **Hair (10 min)**. Empty layer, Clone/Heal with **Sample: All Layers**. Short strokes that follow the hair remove flyaways against the background; clone nearby strands along their flow to fill a thin gap. Leave a few strays.
- 5 **Final review (8 min)**. Fit to screen, **View ▶ Flip Horizontal** (it flips only the view; choose it again to return). Toggle all your Module 3 work off and on. If the before looks untouched and the after just looks like a good day, you've nailed the natural line. Save the finished layered file under a new name (for example **portrait-finished.psd**). Deliver. 🏆

◆ YOUR TURN: MY FINISHING SETTINGS

Record what you actually used, so the numbers become your starting routine.

Eyes: how far I lifted the Curves midtones (describe)	_____
Eyes: High Pass radius / blend mode	_____ px / _____
Teeth: Yellows saturation / Master lightness	_____ / _____
Hair: flyaways removed / strays I deliberately kept	_____ / _____
What the flipped view exposed	_____

◆ YOUR TURN: CONTEXT SETS THE AMOUNT (NOW YOU)

At home: finish the same portrait for a magazine (push a little) and for a LinkedIn headshot (push almost nothing). Note what you changed between the two, and your time for three portraits back to back.

Magazine version

LinkedIn version

Three portraits, timed: _____ min • _____ min • _____ min

◆ MODULE 3 DELIVERABLE CHECKLIST

- ◆ ☐ Even skin from frequency separation (3.1)
- ◆ ☐ Form restored with dodge & burn (3.2)
- ◆ ☐ Eyes brightened and sharpened through a mask
- ◆ ☐ Teeth de-yellowed then gently lifted, masked
- ◆ ☐ Flyaways tamed, a few strays left, on their own layer
- ◆ ☐ Passes the flip-and-mirror test: reads natural, not retouched

◆ SELF - CHECK

Q1

What shape does every finishing move in this lesson share?

- A) Make an adjustment, hide it behind a black mask, and paint it back only where it belongs
- B) Apply one adjustment to the whole image, then lower its opacity
- C) Build a Low and High pair and edit the High layer
- D) Duplicate the photo and edit the pixels directly

Q2

What's the correct order for whitening teeth with a Hue/Saturation layer?

- A) Raise Master lightness first, then reduce Yellows saturation
- B) Raise Master saturation, then lower Master lightness
- C) Push Yellows lightness to maximum
- D) Drop Yellows saturation first, then raise Master lightness slightly

Q3

Which is the lesson's method for sharpening the iris?

- A) Unsharp Mask on the Background layer, unmasked
- B) Merged copy, High Pass at about 1–3 px, blend Overlay or Soft Light, black mask painted white over the iris only
- C) High Pass at about 20 px on a Normal layer over the whole face
- D) The Sharpen tool over the whole eye, catchlight included

Q4

How do you tame flyaway hair the lesson's way?

- A) Remove every stray hair so the outline is perfectly smooth
- B) Clone directly on the photo layer with Sample: Current Layer
- C) On an empty layer, Clone/Heal with Sample: All Layers, short strokes along the hair's direction, leaving a few strays
- D) Blur the edge of the hair until the flyaways disappear

Q5

What's the difference between View ► Flip Horizontal and Image ► Image Rotation ► Flip Canvas Horizontal?

- A) None; both flip the actual pixels
- B) View ► Flip Horizontal flips the pixels; Flip Canvas only flips the view
- C) Only Flip Canvas can be undone
- D) View ► Flip Horizontal flips only the view; Flip Canvas flips the pixels, so you must flip back before saving

Q6

On a real job, in what order do the Module 3 steps go?

A) Eyes/teeth/hair → frequency separation → dodge & burn → global corrections

B) Global corrections → frequency separation → dodge & burn → eyes/teeth/hair

C) Dodge & burn → frequency separation → eyes/teeth/hair → global corrections

D) Frequency separation → global corrections → eyes/teeth/hair → dodge & burn

◆ JOURNAL

Where is your natural line? Describe a finish you have seen (an ad, a friend's photo) that crossed it, and name exactly which move gave it away. Also note: key concepts · what clicked · questions · ideas to try · how you feel about your progress.

◆ NOTES

YOU FINISHED MODULE 3 🎉

You now command the full high-end portrait pipeline: even, textured skin; sculpted light and dimension; and the natural finishing touches that sell it. Just as important, you practiced restraint, the judgment to stop before an edit betrays itself. That taste, more than any single technique, is what clients pay retouchers for. Before Session 4: take one portrait through the complete 3.1 → 3.2 → 3.3 pipeline and keep the layered file, save your finishing layers as a reusable group, and write your journal entries.

Commercial Compositing

An element is only believable when it agrees with the photo it lands in: the same horizon, the same light, the same air, the same grade. You will build one advertising-style key visual across three lessons: place it (4.1), light and anchor it (4.2), then give it depth and one grade (4.3). It is one layered file, so save often and never flatten the master.

Lessons 10-12 (4.1-4.3)

3 lessons

Session 4

Name: _____

HOW TO USE THIS WORKBOOK

Each lesson has **By the end you can**, **Key ideas**, a **Worked example**, a **Your turn** activity (do it in Photoshop and write what happened), a **Self-check**, a **Journal** prompt, and **Notes**. Shortcuts are written **Windows / Mac**. Practice files are in `Labs/module-4/practice/`, or use your own background and a cutout from your Module 1 work. Your working file for all three lessons: `m4-key-visual.psd`.

GROWTH MINDSET: YOUR EYE IS THE TOOL YOU ARE TRAINING

After today you will see horizons, contact shadows, and haze everywhere: in ads, on posters, in the room around you. That is not distraction; it is the skill. Every composite you study, good or bad, now teaches you something, and every "why does this look off?" you can answer is a step toward work nobody can tell was composited.

MODULE PROGRESS TRACKER

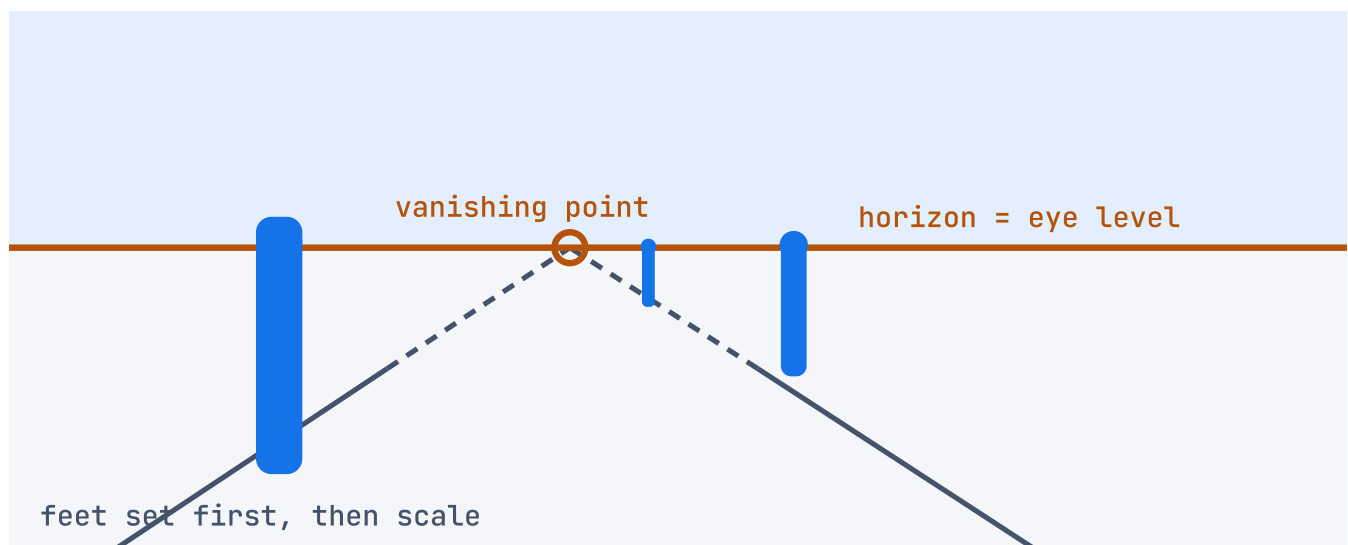
- ◆ ☐ **L10 (4.1)**: my element shares the scene's horizon, stands on the ground at the right size, and its planes run to the vanishing point
- ◆ ☐ **L11 (4.2)**: my element is relit to match, with a tight contact shadow, a graduated cast shadow, and a reflection if the floor is glossy
- ◆ ☐ **L12 (4.3)**: far elements recede with contrast and haze; one grade on top; vignette + grain; key visual exported


◆ BY THE END YOU CAN

- ◆ Find a photo's **horizon line** and read the camera's eye level from it
- ◆ Match an element's eye level and **vanishing points** to the background
- ◆ Scale an element with the **ground-contact rule**, not guesswork
- ◆ Recognize and match a **lens**: wide vs long, and the distortion each brings
- ◆ Correct a mismatch with **Perspective Warp** and **Free Transform**

◆ KEY IDEAS

- ◆ **Every photo has one horizon** at the camera's eye level (the height the lens was at), visible or not. Every added element must share it.
- ◆ **Find it from the scene's own lines.** Receding parallel edges (road edges, window rows, a table's sides) converge on a **vanishing point**, and that point sits on the horizon.
- ◆ **Scale by the ground, not the eye.** For people of similar height on the same flat ground, the horizon crosses each of them at the same body height (the eyes, when the camera was at standing eye level). Set the feet first; the distance then sets the size.
- ◆ **Lens character:** a wide lens (e.g. 24mm) exaggerates depth and bends verticals near the edges; a long lens (e.g. 200mm) compresses depth and flattens features. **Warp fixes geometry, not optics.**



The horizon cuts every standing figure at the eyes, near or far. Distance sets the size.

◆ WORKED EXAMPLE: THE PLAZA PRACTICE FILE

In `l10-plaza-background.jpg` the tile grout, the window rows and the lamp-post line all converge on one point near the middle of the frame. A horizontal guide through that point is the horizon. Check it: the three people already in the plaza (red jacket near, blue coat middle, green top far) all have their eyes on that guide.

- 1 New empty layer. **Line tool** (or Pen) along two receding edges, extended until they meet. Show rulers with `Ctrl+R / ⌘+R`; drag a horizontal guide through the meeting point. (Guides can only be horizontal or vertical, so they can't follow an angled edge.)
- 2 Place the figure. Choose the tile where the shoes touch the ground; that contact point is now fixed.
- 3 `Ctrl+T / ⌘+T`, scale from the feet until the guide crosses the eyes. Scaling is proportional by default in current versions, so **don't hold Shift** (it now switches proportional scaling off).
- 4 Boxes, signs, buildings: Convert to Smart Object, **Edit ▶ Perspective Warp: Layout** mode, draw a quad on each face; **Warp** mode, drag the pins toward the vanishing point. People: a touch of **Free Transform ▶ Distort** for the lean is usually enough.
- 5 Step back. Flatter or rounder than the surroundings? A small Lens Correction or subtle warp. **Save the layered PSD.**

◆ YOUR TURN: PLACE ONE ELEMENT

Place your element (or `l10-figure-cutout.png`, then `l10-product-box-cutout.png`) into your background (or the plaza). Save as `m4-key-visual.psd`. Record:

Two edges I extended to find the vanishing point

Horizon height (y on the ruler)

Where the horizon crosses the people already there

Where I set my element's ground contact

Final scale % (and what surprised me about it)

- ◆ ☐ Horizon found from the scene's own converging lines
- ◆ ☐ Element's ground-contact point placed before scaling
- ◆ ☐ Scaled so the horizon crosses it at the correct body height
- ◆ ☐ Strong planes warped toward the scene's vanishing point
- ◆ ☐ Lens character sanity-checked, no obvious flat/round mismatch

Drill: on `l10-horizon-practice-sheet.jpg`, extend the lines in panels 1–5 and mark each horizon. In panels 3 and 4 both vanishing points fall off the panel; they still sit on one horizontal line. Use panel 6 for one of your own

photos.

◆ WATCH OUT

Perspective Warp grayed out? Turn on Use Graphics Processor in Preferences ► Performance (restart if asked), and select a single layer, ideally a Smart Object. And remember: you can't add depth a long lens removed. If the lenses are wildly different, find a better source photo.

◆ SELF-CHECK

Q1

What does a photo's horizon line represent?

- A) The line where sky meets land, and only when you can see it
- B) The camera's eye level, the height the lens was at, whether or not it is visible
- C) The vertical center of the frame
- D) The line where the sharpest focus falls

Q2

The horizon is hidden behind buildings. How does the lesson find it?

- A) Drag a guide to the exact middle of the image
- B) Run Perspective Warp in Layout mode on the background
- C) Drag guides along the angled edges of the road
- D) Extend two receding edges until they meet, then draw a horizontal guide through that vanishing point

Q3

The scene was shot at standing eye level, with people on flat ground. How do you scale a pasted person?

- A) Set the feet's ground-contact point first, then scale until the horizon crosses their eyes like everyone else's
- B) Scale until it looks about right
- C) Match the pixel height of the nearest person in the scene
- D) Scale until the top of the head touches the horizon

Q4

In current Photoshop versions, what does holding **Shift** while dragging a Free Transform corner do?

- A) Constrains proportions
- B) Scales from the center
- C) Switches proportional scaling off
- D) Nothing; it only matters for rotation

Q5

Edit ► Perspective Warp is grayed out. What does the lesson say to check?

- A) Flatten the image first
- B) Convert the document to 8-bit
- C) Rasterize the Smart Object
- D) Turn on Use Graphics Processor in Preferences ► Performance

Q6

A telephoto headshot pasted onto a wide-angle street still looks subtly wrong after the perspective is matched. What is true?

- A) Perspective Warp can restore the depth the long lens removed
- B) Warp fixes geometry, not optics; a source shot on a focal length close to the background's is the real fix
- C) A Drop Shadow will hide the difference
- D) Scaling it larger cancels the lens compression

◆ JOURNAL

Find a composite ad or movie poster that feels "off." Can you locate the horizon and spot the element whose eye level disagrees with the rest? Also note: key concepts · techniques that clicked · questions · ideas to try · your progress and feelings.

◆ NOTES

◆ BY THE END YOU CAN

- ◆ Read a scene's light: **direction**, **quality** (hard/soft), and **color**
- ◆ Relight an element with **dodge & burn** and masked color so it matches
- ◆ Build a believable **cast shadow** (transform, blur, gradient opacity)
- ◆ Add the tight **contact shadow** that glues feet to the floor
- ◆ Fake a convincing **reflection** on a glossy surface

◆ KEY IDEAS

- ◆ **Read the light first.** Where is it coming from (existing shadows point away from the source)? How hard (crisp shadow edges = hard, direct sun; soft edges = overcast or diffused)? What color (warm sun, cool shade, green from foliage)? The element must obey all three.
- ◆ **Two shadows, two jobs.** The cast shadow tells you where the light is; the contact shadow tells you the object is touching the world. Skip either and something feels wrong.
- ◆ **Real shadows change along their length:** crisp and dark where they touch the object, soft and pale where they end.
- ◆ **Shadows are colored, not black.** Outdoor shadows lean cool/blue (lit by the sky); indoor shadows pick up the room's bounced color.

Subject (cut-out) + clipped D&B, color

Contact shadow • Multiply
tiny, dark, ~2 px blur, right under the base

Cast shadow • Multiply
skewed away from light, blur + fade with distance

Background 

Two shadow layers on Multiply under the subject: the ground texture keeps showing through.

◆ WORKED EXAMPLE: BUILDING THE CAST SHADOW

- 1 **C**tr^L/⌘-click the element's **layer mask thumbnail** (or the layer thumbnail if it has no mask) to load its silhouette.
- 2 New layer **below** the element. **Edit ▶ Fill** with a very dark color, deselect, set the layer to **Multiply**. (Filling a new layer works even for a masked cut-out or a Smart Object.)
- 3 **Free Transform ▶ Distort/Skew** it onto the ground, running away from the light.
- 4 Blur that grows with distance: **Filter ▶ Blur Gallery ▶ Tilt-Shift** or **Field Blur** (sharp pins near the object, strong blur at the far end), or blur a duplicate heavily and blend the two with a gradient mask.
- 5 Gradient mask so it fades out; tint it toward the scene's shadow color (cool/blue outdoors). Match its depth to the real shadows, then usually pull back 5–10%.

◆ YOUR TURN: LIGHT & ANCHOR

Before you touch the element, read your scene (or the plaza, or the two light references) and fill in the table:

Light direction (which way do shadows fall?)
Hardness (crisp or soft shadow edges?)
Light color / shadow color
Which side of my element must get brighter?
Is the ground glossy? (If yes: reflection)

Then: a clipped **50% gray Soft Light** layer (Layer ▶ New ▶ Layer, Mode Soft Light, tick Fill with 50% gray, clip it) to dodge the lit side and burn the other; a clipped **Color Balance** or **Photo Filter** for the light's color; contact shadow; cast shadow; reflection if glossy. Save the layered PSD.

- ◆ ☐ Element re-lit to match the scene's direction, hardness, and color
- ◆ ☐ Tight contact shadow at the base (Multiply)
- ◆ ☐ Cast shadow running away from the light, blurred + faded with distance
- ◆ ☐ Shadows tinted (not pure black) to match the scene
- ◆ ☐ Reflection added if the surface is glossy

Reflection recipe (try it on `l11-glossy-floor-scene.jpg`): duplicate the element, **Edit ▶ Transform ▶ Flip Vertical**, align the bases perfectly, opacity about **25–40%**, Motion Blur or Gaussian Blur that increases downward, mask a fade.

◆ WATCH OUT

The **Drop Shadow** layer style sits at a fixed offset and never lies on the ground in perspective. If you start from it, use Layer ▶ Layer Style ▶ Create Layer to split it onto its own layer, then distort, blur, and mask it like a hand-built shadow.

Q1

The shadows in your background have crisp, sharp edges. What does that tell you about the light?

- A) It is overcast or diffused
- B) It is warm-colored
- C) It is a hard, direct source, such as the sun
- D) It is behind the camera

Q2

Why are the contact and cast shadow layers set to Multiply?

- A) So the ground texture shows through and the shadow darkens the floor instead of painting a flat gray blob
- B) Because Multiply blurs the layer automatically
- C) Because Free Transform ► Distort only works on Multiply layers
- D) So the shadow turns cool automatically

Q3

Your cut-out uses a layer mask. How does the lesson create the cast-shadow shape?

- A) Lock Transparent Pixels on the element and fill it black
- B) Apply the Drop Shadow layer style and leave it as is
- C) Duplicate the element and drag Brightness to -100
- D) Ctrl/⌘-click the mask thumbnail, fill the selection with a very dark color on a new layer below, deselect, set Multiply

Q4

How should a believable cast shadow change as it travels away from the object?

- A) It stays equally dark and sharp all the way
- B) Crisp and dark near the object, softer and paler toward its end
- C) Softer near the object, sharper at the tip
- D) Darkest at the tip

Q5

What color should an outdoor shadow lean toward?

- A) Cool/blue, because the sky, not the sun, lights it
- B) Pure black
- C) The color of the object casting it
- D) Warm orange, like the sun

Q6

Which is the lesson's reflection recipe for a glossy floor?

- A) A Drop Shadow at 100% opacity, offset downward
- B) Duplicate, Flip Horizontal, keep it at full opacity
- C) Duplicate, Flip Vertical, align the bases, lower opacity to ~25–40%, blur more with distance, mask a fade
- D) Paint it by hand with the Mixer Brush

◆ JOURNAL

Look around the room and study one object's shadow. Where is the crisp contact shadow, where does the cast shadow soften, and what color is it? Retelling it in words trains your compositing eye. Also note: key concepts · techniques that clicked · questions · ideas to try · your progress and feelings.

◆ NOTES

◆ BY THE END YOU CAN

- ◆ Use **atmospheric perspective** (haze, fading contrast) to build depth
- ◆ Add fog, light rays, and dust that sit **between** elements, not just on top
- ◆ Lay a single **unifying grade** over the whole stack
- ◆ Explain why one grade over everything beats grading each piece
- ◆ Deliver a finished, cohesive **key visual**

◆ KEY IDEAS

- ◆ **Distance has a look.** The farther something is, the more air washes it out: contrast drops, colors desaturate and shift toward the sky's color (usually cooler and lighter), edges soften.
- ◆ A background element as crisp and saturated as the foreground feels **pasted forward**. Push it back with lower contrast and a little haze.
- ◆ **Atmosphere goes between, grade goes on top.** Haze sits above the far elements and below the subject and its shadows; the grade sits at the very top and rains down on everything.
- ◆ **Unify last, grade once.** Get every piece neutral and correct first, then one grade. One dial beats twenty.

★ Unifying Grade (LUT / Curves)

on top of
everything

Subject + clipped light/color

Contact + cast shadows • Multiply

Atmosphere / haze • low opacity
over the far elements only

between

Background 

The grade rains down on everything below it, atmosphere included; that shared color is what unifies.

◆ WORKED EXAMPLE: THE FOUR FINISHING STEPS

STEP	WHAT YOU DO	WHY
1 · Depth by contrast	Clipped Curves on far elements: lower contrast, lift the blacks slightly, a touch of sky color. Nearest element stays crispest.	The contrast falloff alone reads as distance
2 · Atmosphere	New layer (Normal), soft brush, scene/sky color, very low flow, glazed over the far elements. Optional rays or particles on Screen.	Fills the space between elements: a world, not stacked layers
3 · Unifying grade	At the top: a Color Lookup, or a Curves + Color Balance group (cool shadows, warm highlights, gentle contrast). Pull opacity back.	One shared color = one photograph
4 · Finish & deliver	Subtle vignette, a touch of glow, grain. Flatten a copy, convert to sRGB, export.	Holds the eye; binds everything to one "film stock"

◆ YOUR TURN: DEPTH & GRADE (MODULE 4 DELIVERABLE)

Label your scene's depth bands before you add a single layer. On the plaza, place `l12-far-clocktower-cutout.png` on the distant hills first; it arrives far too crisp.

My far element(s)

My mid element(s)

My near element (crispest, most saturated)

Haze color I sampled (from where?)

My grade in words (shadows pushed to... highlights to...)

- ◆ ☐ Element placed with correct perspective & scale (4.1)
- ◆ ☐ Element lit, with contact and cast shadows (4.2)
- ◆ ☐ Depth set by lowering contrast / adding haze on far elements
- ◆ ☐ Atmosphere (haze, optional rays/particles) between elements
- ◆ ☐ One unifying grade over the whole stack
- ◆ ☐ Vignette + grain; exported as a finished key visual 🏆

◆ WATCH OUT

A haze layer placed **above** the grade escapes it and looks pasted on. And if the composite still looks "off," run the checklist in order: horizon, scale, lens, light direction, shadow, contact shadow, depth/haze, grade. Ninety percent of "off" composites fail at scale or the missing contact shadow; fix those before blaming the grade.

Q1

Atmospheric perspective: what happens to things as they get farther away?

- A) They get darker and more saturated
- B) Their edges get sharper
- C) They get warmer and higher in contrast
- D) Contrast drops, colors desaturate and shift toward the sky color, and edges soften

Q2

Where does the depth-haze layer sit in the stack?

- A) At the very top, above the grade
- B) Above the far elements, below the subject, its shadows, and the grade
- C) Clipped to the subject
- D) Below the background layer

Q3

Why does one grade over the whole stack beat grading each element separately?

- A) It uses less memory
- B) Color Lookup layers only work at the top of a stack
- C) A single set of adjustments touches every element, so they share one color identity, and you tune one dial instead of twenty
- D) It removes the need for shadows

Q4

What should be true of every piece before you add the unifying grade?

- A) It is neutral and correct
- B) It is flattened into the background
- C) It is converted to 8-bit
- D) It already has a vignette

Q5

Which of these can serve as the unifying grade, according to the lesson?

- A) A Drop Shadow layer style on the subject
- B) A Color Lookup (LUT), a low-opacity Gradient Map, or a Curves + Color Balance group at the top
- C) A Hue/Saturation layer clipped to the subject
- D) A Gaussian Blur on the background

Q6

How does the lesson finish and deliver the key visual?

- A) Flatten the master and save over the PSD
- B) Export the layered PSD straight to the web
- C) Take a screenshot of the canvas
- D) Vignette, glow and grain, then flatten a copy, convert to sRGB, and export

◆ JOURNAL

Pick a film still you love. Describe its grade in words (where are the shadows pushed, where the highlights?) and where atmosphere creates its depth. Then try to rebuild that grade on one of your own images. Also note: key concepts · techniques that clicked · questions · ideas to try · your progress and feelings.

◆ NOTES

MODULE 4 SUMMARY

- ◆ **Perspective first:** share one horizon, place by ground contact, match the lens.
- ◆ **Light & shadow next:** match direction, hardness and color, then anchor with a contact shadow and a cast shadow.
- ◆ **Atmosphere builds depth:** lower contrast and add haze on far elements.
- ◆ **One unifying grade** over the whole stack makes separate photos read as one frame.

BEFORE THE NEXT SESSION

- ◆ ☐ Find the horizon in five photos by extending their receding lines (use panel 6 of the drill sheet for one)
- ◆ ☐ Study the contact and cast shadows of three real objects around you
- ◆ ☐ Finish the solo variations you started in class
- ◆ ☐ Complete one composite through the full 4.1 → 4.2 → 4.3 pipeline and keep the layered file
- ◆ ☐ Save your unifying grade as a reusable LUT (File ► Export ► Color Lookup Tables, Lesson 2.2)
- ◆ ☐ Write your three Learning Journal entries

YOU FINISHED MODULE 4 🎉

You can now take elements shot at different times, in different light, on different lenses, and assemble them into one believable image: matched in geometry, married in light and shadow, and unified by atmosphere and grade. Commercial compositing is one of the highest-paid Photoshop skills, and you now own its full pipeline. Next session: Module 5, Design Systems & Type.

Design Systems & Type

From one good image to a small production system. You will build a type system you can change in one place, a mockup template you can refill forever, and a full campaign (poster, square, story, banner) from a single master file.

Lessons 13–15 (5.1–5.3)

3 lessons

Session 5

Name: _____

HOW TO USE THIS WORKBOOK

Each lesson has **By the end you can**, **Key ideas**, a **Worked example** (the guided build), a **Your turn** activity, a **Self-check**, a **Journal** prompt and **Notes**. Shortcuts are written **Windows / Mac**. Practice files are in `labs/module-5/practice/`. Before you start: make one project folder (for example `northwind-campaign/`) and keep everything from today in it. Linked files break if you move them.

GROWTH MINDSET: TASTE IS A SET OF DECISIONS

"Good taste" in type and layout isn't magic; it's decisions made once and applied consistently. A scale, a pair of typefaces, a grid: each is a rule you can name. Templates feel like extra work the first time and pure magic every time after. Expect the first build to feel slow. That's the setup cost of a system, and it pays you back on every job.

MODULE PROGRESS TRACKER

- ◆ ☐ **L13:** a written scale, named Paragraph/Character Styles, columns + a grid at the body leading; one style edit updates the page
- ◆ ☐ **L14:** a warped Smart-Object mockup with clipped Multiply/Screen layers; art swapped with Replace Contents; a linked logo updated from its source
- ◆ ☐ **L15:** one master with four artboards, linked art, shared styles, exported in one pass; one source edit updates the set 🏆

◆ BY THE END YOU CAN

- ◆ Build a **modular type scale** and a clear hierarchy from it
- ◆ Create and apply **Paragraph and Character Styles**
- ◆ Turn on real **OpenType**: ligatures, true small caps, oldstyle figures
- ◆ Set up **columns** and a **baseline grid** (the document grid, spaced at your leading)
- ◆ Pair typefaces from **Adobe Fonts** with intent

◆ KEY IDEAS

- ◆ **Modular scale**: one base size × one ratio (1.25, 1.333, 1.618). Assign roles to steps and never freehand a size again.
- ◆ **Four cheap levers of hierarchy**: size, weight, color and space. Often weight or space says "this matters" better than size.
- ◆ **Styles bottle a spec** (font, size, leading, tracking, color, OpenType) under a name. Edit the style once and every instance updates.
- ◆ **No faux styling**. Faux Bold and Faux Italic smear a regular weight. Use a real Bold/Italic or choose another face.
- ◆ **Photoshop has no true baseline grid**. Set the document grid to your body leading, give text that leading (or a multiple), and align each block's first baseline by eye.
- ◆ **Usually two typefaces**: one display, one text, chosen to contrast.

Display 39

H1 31

× 1.25

H2 25

× 1.25

Subhead 20

× 1.25

Body 16: the reading size it all tunes to

× 1.25

Caption 13

× 1.25

one base size × one ratio → a scale where every pairing already agrees

A modular scale from a 16 px base and a 1.25 ratio. Assign roles to steps; the math guarantees the sizes relate.

YOU WANT...	WHERE IT LIVES
Type sized in pixels	Preferences ► Units & Rulers ► Type ► Pixels
Columns, gutters, margins	View ► Guides ► New Guide Layout (older versions: View ► New Guide Layout)
A baseline grid (workaround)	Preferences ► Guides, Grid & Slices: Gridline Every = body leading, Subdivisions 1; View ► Show ► Grid (Ct r l / ⌘ + ')
Named styles	Window ► Paragraph Styles and Window ► Character Styles (two panels)
Ligatures, small caps, oldstyle figures	Character panel: OpenType buttons along the bottom (or panel menu ► OpenType)

◆ WORKED EXAMPLE: A TYPE SYSTEM (THE GUIDED BUILD)

- 1 Set the scale.** Units & Rulers ► Type ► Pixels (or keep points and use the same numbers as pt). Base 16, ratio 1.25 → 13 / 16 / 20 / 25 / 31 / 39. Caption 13, Body 16, Subhead 20, H2 25, H1 31, Display 39.
- 2 Build the grid.** New Guide Layout: 3 or 6 columns, a gutter, margins. Then Preferences ► Guides, Grid & Slices: Gridline Every 24 px (for 16 px body), Subdivisions 1. Show it with Ct r l / ⌘ + '.
- 3 Create styles.** Set one line per role correctly (font, size from the scale, leading = grid increment or a multiple, tracking, color, OpenType on). With it selected, create a new Paragraph Style and name it. Repeat per role; add Character Styles for links/emphasis.
- 4 Prove the system.** Lay out the page with named styles only. Nudge each block so its first baseline sits on a gridline (zoom in, arrow keys). Edit the Body style by one step and watch the page re-flow. Save the PSD: its styles feed Lesson 5.3.

◆ YOUR TURN: WRITE THE SYSTEM BEFORE YOU BUILD IT

Use the copy in `l13-layout-copy.txt`. Fill in your scale (lesson numbers, or ×3 for an 1800 × 2400 px practice poster):

ROLE	SIZE	LEADING	FONT & WEIGHT	STYLE NAME
Display				
H1				
H2				
Subhead				

ROLE

SIZE

LEADING

FONT & WEIGHT

STYLE NAME

Body

Caption

Base _____ × ratio _____ · Grid (Gridline Every) _____ · Columns _____ gutter _____

Diagnose `113-ransom-note-before.png` : list every inconsistency you can find (fonts, sizes, alignments, colors, faux styling).

- ◆ ☐ A written modular scale with roles assigned
- ◆ ☐ Columns via New Guide Layout + a document grid set to your body leading
- ◆ ☐ Paragraph & Character Styles created and named
- ◆ ☐ Real OpenType on; no faux bold/italic
- ◆ ☐ Editing one style updates every instance

◆ WATCH OUT

Text won't snap to the grid? It never will: View ► Snap To ► Grid snaps a layer's **edges**, not its baselines. Keep leading equal to the grid increment and align each block's first baseline by eye. Style not updating one block? You changed that block by hand; clear the override in the Paragraph Styles panel.

◆ SELF-CHECK

Q1

With a 16 px base and a 1.25 ratio, which set of sizes does the lesson's modular scale give you?

A) 12 / 16 / 20 / 24 / 28 / 32

B) 14 / 16 / 18 / 20 / 22 / 24

C) 13 / 16 / 20 / 25 / 31 / 39

D) 16 / 32 / 48 / 64 / 80 / 96

Q2

The client wants all body text one step larger across a styled layout. What makes this a single edit?

- A) Editing the Body Paragraph Style once, so every instance updates
- B) Selecting every text layer and typing a new size
- C) Scaling the whole document with Image Size
- D) Converting all text to a Smart Object

Q3

How do you build a baseline grid in Photoshop?

- A) View ► Guides ► New Guide Layout, Rows = body leading
- B) Turn on Snap To Baseline in the Paragraph panel
- C) Photoshop draws one automatically when you create a Paragraph Style
- D) Preferences ► Guides, Grid & Slices: Gridline Every = your body leading, Subdivisions 1

Q4

Snap To ► Grid is on, but your text block's baselines still don't sit on the gridlines. Why, and what do you do?

- A) The grid is broken; restart Photoshop
- B) Snapping works on layer edges, not baselines; zoom in and align the first baseline by eye with the arrow keys
- C) Convert the text to shapes so it can snap
- D) Set the leading to Auto so it matches the grid

Q5

Your chosen typeface has no Bold weight, and your H1 style needs one. What does the lesson say to do?

A) Click the Faux Bold button in the Character panel

B) Add a thin Stroke layer style in the same color

C) Choose a different face that ships a real Bold

D) Increase the tracking until it looks heavier

Q6

What does View ► Guides ► New Guide Layout create?

A) Columns and rows, with gutters and margins, but not a baseline grid

B) A true baseline grid that text snaps to

C) A set of named Paragraph Styles

D) Artboards at preset sizes

◆ JOURNAL

Find two typefaces you think pair well and articulate why — contrast in structure, shared era, matching x-height? Naming the reason turns taste into a repeatable skill. Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

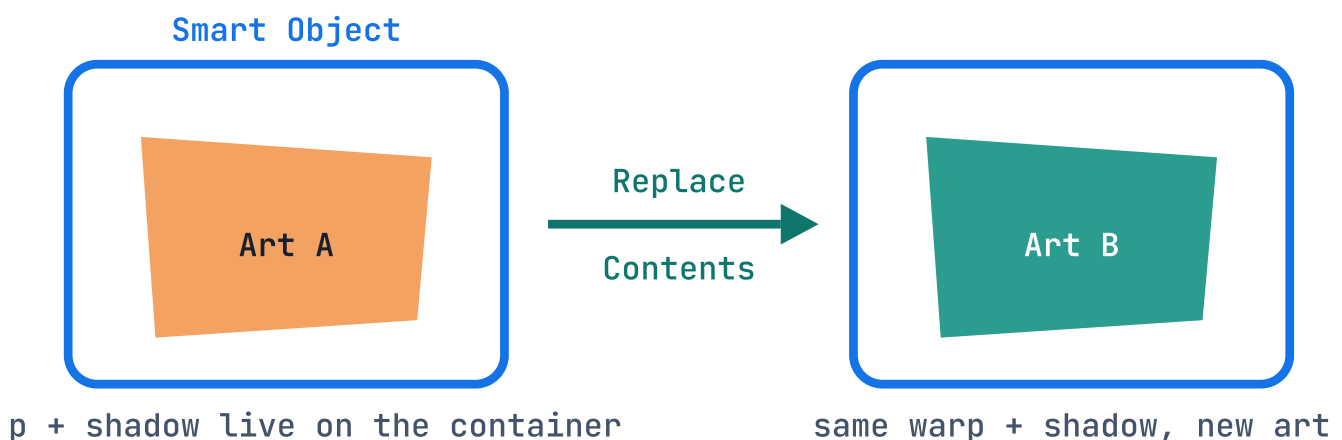
◆ NOTES

◆ BY THE END YOU CAN

- ◆ Explain what a **Smart Object** protects and why it is non-destructive
- ◆ Choose between **embedded** and **linked** Smart Objects
- ◆ Build a **mockup template** with a warped placeholder
- ◆ Use **Replace Contents** and **Place Linked** to swap art in one move
- ◆ Update **one linked file** and have every design using it refresh

◆ KEY IDEAS

- ◆ **A container you can refill.** Transforms, warps, Smart Filters and layer styles live on the Smart Object; Replace Contents swaps what's inside and keeps all of them.
- ◆ **The catch:** the container's transform is applied to whatever you swap in. Build placeholders at the exact pixel size (or at least the exact aspect ratio) of the art they'll hold.
- ◆ **Embedded** = stored inside the PSD, each copy independent (one-off protection). **Linked** (File ► Place Linked) = points at a file on disk that many documents share (the backbone of a system).
- ◆ **Effects on the container, not the art.** Warp the placeholder; put shadows on a Multiply layer and highlights on a Screen layer above it, clipped.
- ◆ **Link discipline.** Don't move or rename linked files; File ► Package gathers them for handoff.



The container holds the treatment; the contents are swappable.

◆ WORKED EXAMPLE: A REUSABLE MOCKUP (THE GUIDED BUILD)

- 1 **Place the placeholder.** Open `l14-mockup-poster-wall.jpg` (or your own surface photo). Rectangle tool (U), click once and enter the art's exact size: **1200 × 1800 px** for the practice designs.

Right-click ► Convert to Smart Object. Free Transform (Ctrl/⌘+T) ► Warp (or Perspective Warp) to fit the surface.

- 2 **Light it into the scene.** A layer set to **Multiply** above the placeholder: paint the surface's shadows and creases back over the art. A **Screen** layer for highlights. Clip both to the placeholder (Alt/⌘-click the line between the layers).
- 3 **Swap in real art.** Double-click the placeholder (it opens in its own tab), place your design, save, close. Prefer Place Linked for art that several mockups share.
- 4 **Prove reuse.** Layer ► Smart Objects ► Replace Contents with another design at the same pixel size. The perspective, shadows and highlights hold. Save as your template.

◆ YOUR TURN: SWAP TEST AND LINK TEST

Swap each practice design into your placeholder and record what happens:

DESIGN	PIXEL SIZE	HOW I SWAPPED IT	DID THE WARP AND LIGHTING HOLD? DID IT FIT?
l14-art-a-sunrise	1200 × 1800		
l14-art-b-geometric	1200 × 1800		
l14-art-c-typographic	1200 × 1800		
l14-art-d-landscape	1800 × 1200		

Link test: copy `l14-Logo-v1.png` into your project folder as `Logo.png`. In a new document, File ► Place Linked it twice. Change `Logo.png` (edit its single layer and save, or replace it with `l14-Logo-v2.png` under the same name). What happened in the document?

- ◆ ☐ Placeholder is a Smart Object, warped to the surface
- ◆ ☐ Shadows (Multiply) + highlights (Screen) on the container, clipped
- ◆ ☐ Art swaps via Replace Contents / double-click edit
- ◆ ☐ Shared art uses Place Linked where appropriate
- ◆ ☐ Saved as a reusable template; Package if handing off

◆ WATCH OUT

New art came in too big or squashed? It has different pixel dimensions from the placeholder. Export it at the placeholder's size, or double-click the placeholder and paste it into the existing canvas. A linked layer shows a warning? The file moved or was renamed: right-click ► Relink to File, and stop moving it.

◆ SELF-CHECK

Q1

You warped a Smart-Object placeholder and added clipped shadow layers. What happens to that treatment when you use Replace Contents?

- A) It is removed, and you rebuild it for the new art
- B) It stays on the container and applies to the new art
- C) It is baked permanently into the old art
- D) Only the layer styles survive; the warp resets

Q2

What is the key difference between an embedded and a linked Smart Object?

- A) Embedded objects can be scaled without loss; linked ones cannot
- B) Linked objects cannot take Smart Filters
- C) Embedded objects update automatically when the source file changes
- D) A linked one points to one file on disk that many documents can share; an embedded one is stored inside the PSD

Q3

After Replace Contents, the new design lands too big and misaligned. What's the most likely cause?

- A) The new art has different pixel dimensions, and the container's transform is applied to it
- B) The document is in the wrong color mode
- C) The shadow layers aren't clipped
- D) Replace Contents only works with PSB files

Q4

Where should a mockup's shadows and highlights go?

- A) Painted directly into the artwork before placing it
- B) Inside the Smart Object, on top of the art
- C) On Multiply and Screen layers above the placeholder, clipped to it
- D) As a Drop Shadow layer style on the Background layer

Q5

You're sending a project with linked assets to a colleague. What should you do first?

- A) Rasterize every Smart Object
- B) File ► Package, which collects the linked files with the document
- C) Rename the linked files so they're easier to find
- D) Flatten the image

Q6

You tidied your desktop and moved `Logo.png` into a new folder. What happens in the documents that link it?

- A) Nothing; Photoshop tracks the file automatically
- B) The logo is converted to an embedded copy
- C) The logo layer is deleted
- D) Photoshop loses the link and the layer shows a warning until you relink it

◆ JOURNAL

Think of a repetitive design task you have done by hand. How would a Smart-Object template and a linked asset have collapsed it into a single edit? Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

◆ NOTES

◆ BY THE END YOU CAN

- ◆ Use **artboards** to hold multiple sizes in one document
- ◆ Share art across artboards with **linked Smart Objects**
- ◆ Adapt one design to portrait, square and landscape **without redrawing it**
- ◆ Export every artboard at once with **Export As**
- ◆ Deliver a coherent, on-brand **campaign set** from one file

◆ KEY IDEAS

- ◆ **Master and instances.** The master owns the shared truth (key art, logo, tagline, grade); each artboard is a view onto it at one size and crop. Only layout and framing differ.
- ◆ **Design at the biggest size, export down.** Usually the print poster. Scaling a small master up softens everything.
- ◆ **One color mode, one resolution** for every artboard in a document. Keep the master RGB; CMYK for print is a separate final step on an exported copy.
- ◆ **Recompose, don't squash.** Anchor the focal point first; stack for the story, center for the square, spread for the banner. Respect safe zones.
- ◆ **Adjustment layers can't be linked.** For a shared grade, duplicate the layer (or reload the same LUT) into each artboard and update each copy.

ARTBOARD	SIZE (PX)	SHAPE	COMPOSITION
Poster	3300 × 5100	Portrait	Design first, at full resolution
IG square	1080 × 1080	1:1	Center
Story	1080 × 1920	9:16	Stack vertically; clear of UI overlays
Banner	1500 × 500	3:1	Spread horizontally; clear of crop edges

◆ WORKED EXAMPLE: THE CAMPAIGN (THE GUIDED BUILD)

- 1 **Set up artboards.** New document (RGB), Artboards on. Add poster, IG square (1080²), story (1080 × 1920) and banner (1500 × 500), using presets where they exist and custom sizes otherwise. Design the poster first.
- 2 **Share the essentials.** File ► Place Linked the logo and key art into each artboard. Copy a styled text layer in from your L13 file so its Paragraph Styles come along; apply them to each headline and

subhead.

- 3 **Recompose per size.** Anchor the focal point, rearrange around the aspect ratio, respect safe zones. Optional shared grade: the same adjustment layer or the same LUT in a Color Lookup layer at the top of each artboard.
- 4 **Export the set.** File ► Export ► Export As, all artboards, format and scale, Export (or File ► Export ► Artboards to Files). Then open the linked logo or key art, change it visibly, save; every artboard updates (else Layer ► Smart Objects ► Update All Modified Content). Re-export. Save the master with its linked files in the same folder.

◆ YOUR TURN: PLAN EACH FRAME

Use `l15-campaign-copy.txt`. Sketch each artboard's layout before you build: where is the focal point, and what drops out at small sizes?

ARTBOARD	FOCAL POINT ANCHORED AT...	TEXT KEPT	TEXT DROPPED
Poster			
IG square			
Story			
Banner			

My one-edit proof: I changed _____ in _____ and _____ of 4 artboards updated.

- ◆ ☐ One master PSD with poster, square, story and banner artboards
- ◆ ☐ Logo & key art placed as linked Smart Objects (shared source)
- ◆ ☐ Type driven by the 5.1 Paragraph/Character Styles
- ◆ ☐ Each size recomposed for its aspect ratio, safe zones respected
- ◆ ☐ All artboards exported in one pass via Export As
- ◆ ☐ A single source edit updates the whole set 🏆

◆ WATCH OUT

Key art looking soft on the poster? It was scaled past 100%. Use art at or above the poster's size (the practice `l15-key-art.png` is 2400 px wide, so keep it at or below 100% there). Printer asks for CMYK? Don't convert the master: artboards share one color mode. Convert an exported copy as the final step.

◆ SELF-CHECK

Q1

Your campaign has a print poster, a square, a story and a banner. Which do you design first, and why?

- A) The print poster, the largest, so smaller sizes scale down from it
- B) The square, because it's the most common size
- C) The banner, because it's the hardest to fit
- D) The order doesn't matter with artboards

Q2

How should the logo get onto all four artboards so one edit updates every one?

- A) Copy and paste the logo pixels onto each artboard
- B) File ► Place Embedded into each artboard
- C) File ► Place Linked into each artboard (so every copy points to the same file)
- D) Draw it once on the poster and let Export As add it to the others

Q3

What renders every artboard to its own file in one pass?

- A) File ► Save As, once per artboard
- B) File ► Export ► Export As with all artboards (or File ► Export ► Artboards to Files)
- C) Image ► Duplicate
- D) File ► Package

Q4

How do you adapt the poster design to the 3:1 banner?

- A) Scale the whole poster down until it fits the height
- B) Stretch the poster horizontally to 3:1
- C) Crop a thin strip out of the middle of the poster
- D) Anchor the focal point first, then spread the elements horizontally around it

Q5

The printer needs the poster in CMYK, but the social sizes must stay RGB. What's the right approach?

- A) Keep the master RGB and make the CMYK conversion a separate, final step on an exported copy
- B) Set the poster artboard to CMYK and the others to RGB
- C) Convert the whole master to CMYK before exporting anything
- D) Export As converts to CMYK automatically for large artboards

Q6

You edited and saved the linked logo file, but one artboard still shows the old version. What do you do?

- A) Place the logo again on that artboard
- B) Close and delete the master, then rebuild it
- C) Layer ► Smart Objects ► Update All Modified Content
- D) File ► Package

◆ JOURNAL

Find a brand you admire and list every place its campaign appears (feed, story, print, web). What stays identical across them, and what is recomposed? That is the master-and-instances split in the wild. Also note: key concepts · what clicked · questions · ideas to try · your progress and feelings.

◆ NOTES

MODULE 5 SUMMARY

- ◆ A **type system** (scale + styles + grid) gives every layout one consistent voice.
- ◆ **Smart Objects** protect art and, via **linked files**, let one source update many designs.
- ◆ **Artboards** hold every size in one master; **Export As** ships them together.
- ◆ Design the **largest first**, share the essentials, **recompose** per aspect ratio.

YOU FINISHED MODULE 5 🎉

You moved from making one good image to running a small production system: a named type voice (5.1), self-updating templates (5.2), and a full campaign from a single master (5.3). Before Session 6: ship one campaign set (4+ sizes) from a single master, prove a one-edit update, and write your journal entries. Next: Module 6, Generative + Automation, Advanced.

Generative + Automation, Advanced

Today you add speed and scale to your craft. Generative tools take the tedious pixels of a composite; conditional Actions take the repeated steps; Variables and a CSV take the repeated designs; and a short script shows you where automation goes next. The module deliverable is a folder of personalized graphics rendered from one template.

Lessons 16–18 (6.1–6.3)

3 lessons • Session 6

Name: _____

HOW TO USE THIS WORKBOOK

Each lesson has **By the end you can**, **Key ideas**, a **Worked example** (the lesson's Guided Build), a **Your turn** checklist with space to write, a **Recipe card** to keep, a **Self-check** quiz, and a **Journal** prompt. Journal after every lesson: key concepts · what clicked · questions · ideas to try · your progress and feelings.

GENERATIVE FEATURES NEED FOUR THINGS

Generative Expand, Generative Fill, and Harmonize (L16) run in Adobe's cloud. You need **Photoshop 2025 or later**, **an internet connection**, to be **signed in to your Adobe account**, and **generative credits** (every Generate spends some). If they are not available today, your instructor will show the same workflow with offline tools (Crop with Fill: Content-Aware Fill, the Remove tool, and matching light by hand as in Lesson 4.2). L17 and L18 work completely offline.

MODULE PROGRESS TRACKER

- ◆ ☐ **L16:** I gave a cramped composite room, removed a distraction, grew connective tissue, harmonized the subject, and finished by hand
- ◆ ☐ **L17:** I built a delivery Action with a landscape/portrait conditional, batched a folder hands-off, and made a droplet
- ◆ ☐ **L18:** I bound a template to a CSV and exported one graphic per row (**Module 6 deliverable**)
- ◆ ☐ **L18:** I read the sample script, changed a value, and ran it on a test folder

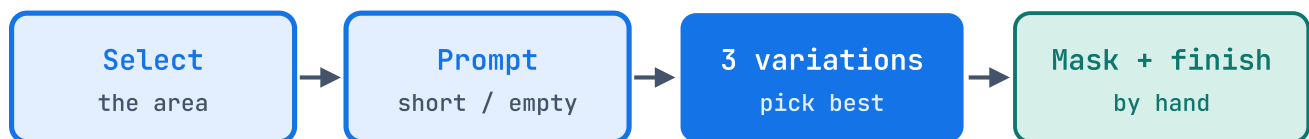
◆ BY THE END YOU CAN

- ◆ Frame generative fill as a **compositing tool**, not a replacement for craft
- ◆ Extend a background with **Generative Expand** to fit a layout
- ◆ Remove distractions and fill gaps with **Generative Fill** + a prompt
- ◆ Generate **connective elements** (shadow ground, foliage, debris) to sell a composite
- ◆ Use **Harmonize** to match a placed element's light and color to its background
- ◆ Judge, mask, and hand-finish generative output, and respect its **limits and licensing**

◆ KEY IDEAS

- ◆ **A fast, fallible assistant.** Like a talented but literal intern: superb at plausible texture (sky, grass, pavement, walls) and removals, unreliable at specifics (hands, text, exact brands, precise counts).
- ◆ **The loop is never "generate and ship."** It is generate → pick the best of three → mask and blend → hand-fix the tells.
- ◆ **Every result lands on a new generative layer**, with variations in the Properties panel. Regenerate as often as you like; each set is kept. Never flatten until the seams are blended.
- ◆ **Short prompts, small selections.** "grass", "clear blue sky", or empty to match. A whole complex scene in one go makes it flail.
- ◆ **Harmonize** (Photoshop 2025 and later) generates versions of a placed element with its light, color, and shadows adjusted to the background. Compare, keep the best, mask back what it shouldn't change.
- ◆ **It won't fix a weak composite.** Mismatched perspective, light, or scale (Module 4) are still yours to fix. Generation fills texture; it doesn't reason about your scene.

raw material on a non-destructive layer • you finish it



The loop is always generate → choose → refine.

◆ WORKED EXAMPLE: HARMONIZE A COMPOSITE

Open a cramped or messy composite: yours from Module 4, or the practice parts (`l16-cramped-scene.jpg`) with `l16-pasted-subject.png` placed hard against the right edge, as in `l16-cramped-composite-`

preview.jpg). Save as `l16-composite-working.psd` .

- 1 **Give it room.** Crop tool (C), options bar **Fill: Generative Expand**. Drag the canvas edges outward, commit, leave the prompt empty, click **Generate**. Pick the cleanest variation in the Properties panel. Check the new edges at 100% (Ct r l / % + 1) for repeats or seams. Tip: hide the pasted subject while you expand, so the model continues the scene, not the person.
- 2 **Remove distractions.** Lasso a distracting object with a little margin (include its shadow). **Generative Fill** with an empty prompt. If a variation leaves a smudge, regenerate or clean it with the Healing Brush.
- 3 **Grow connective tissue.** Select ground beneath the pasted subject and prompt something like `gravel with small shadows` , or generate foreground foliage to overlap an edge. Mask each generative layer so it only contributes where you want.
- 4 **Harmonize the subject.** If it still looks lit by a different photo, select its layer and run **Harmonize** (contextual task bar or Properties panel). Compare the variations with the original; keep the best; mask back wherever it changed something it shouldn't (an edge, a face, a logo).
- 5 **Finish by hand.** Matching grain and a touch of sharpening on the generated regions; dodge & burn to marry their light to the scene; one unifying grade over everything (Lesson 4.3).

◆ YOUR TURN: THE PROJECT CHECKLIST

- ◆ ☐ Canvas extended with Generative Expand, edges audited
- ◆ ☐ A distraction removed cleanly with Generative Fill
- ◆ ☐ Connective element generated and masked in
- ◆ ☐ Subject harmonized to the scene (or light matched by hand), with unwanted changes masked out
- ◆ ☐ Generated regions grain/sharpen-matched and hand-blended
- ◆ ☐ One unifying grade over the whole result

My prompt log. Each prompt, the selection it went into, and what happened (keep the ones that worked):

Harmonize: what did it get right, and what did I mask back to the original?

◆ NOW YOU: SOLO VARIATION (START IN CLASS, FINISH AT HOME)

- ◆ ☐ A phone snapshot turned into a wide banner purely with Generative Expand, every seam fixed by hand until it is undetectable
- ◆ ☐ Three increasingly hard distractions removed (a sign, a person, a shadow across the subject; practice file `l16-three-distractions.jpg`)
- ◆ ☐ The same fill generated five times, and the differences studied

Where did the AI succeed, and where did I have to take over?

◆ WATCH OUT

- ◆ **Grayed out or missing?** Update Photoshop, sign in, check the network and your credits. Harmonize needs a single pixel or Smart Object layer above a background; its location can shift between releases (search Help ► Photoshop Help or the Discover panel).
- ◆ **Softer than the rest?** Generated pixels often carry less fine detail than your capture. Match the grain and add a little sharpening.
- ◆ **Faces, hands, text, logos, product details, repeat patterns** drift. Artifacts may show only at 100%: zoom in and audit.
- ◆ **Licensing.** Firefly models are designed to be commercially safe, but current versions can also offer third-party partner models with different terms. For client work, check which model you used and confirm the terms.

◆ RECIPE CARD: GENERATIVE ASSIST

- 1 Generative Expand (Crop out + empty prompt) for room
- 2 Generative Fill + empty prompt to remove distractions
- 3 Short prompts, small selections for connective texture
- 4 Harmonize a placed subject to the scene, then mask back what it shouldn't change
- 5 Keep each result on its generative layer; mask, don't flatten
- 6 Grain/sharpen-match, hand-blend, then one unifying grade

Mantra: the model gives you a head start, never the finish line.

A NOTE ON HONESTY

For journalism or documentary work, generative changes may be off-limits. Disclose edits when integrity matters. It is a tool, like the Healing Brush; the judgment of what to keep, blend, and reject is still yours.

◆ SELF-CHECK

Q1

Which describes the right way to work with generative tools in a composite?

- A) Generate once, flatten, and export while the result is fresh
- B) Describe the whole finished scene in one detailed prompt
- C) Select, prompt short or empty, pick the best variation, then mask and finish by hand
- D) Generate on the Background layer so the new pixels match the original exactly

Q2

Your composite is framed too tight and needs more room on both sides. What do you do?

- A) Crop tool with Fill: Generative Expand, drag the edges outward, commit, Generate
- B) Free Transform the background layer wider
- C) Image ► Image Size with a larger width
- D) Select the whole canvas and prompt "make it wider"

Q3

A SALE sign on a post casts a shadow across the ground behind your subject. What is the best way to remove it?

- A) Type "remove the sign" as the prompt
- B) Select the sign tightly along its edges and Generate with an empty prompt
- C) Paint over it on the Background layer with the Eraser
- D) Lasso the sign, its post, and its shadow with a little margin, empty prompt, Generate

Q4

What is Harmonize for?

- A) Removing distractions without a prompt
- B) Matching a placed element's light, color, and shadows to its background
- C) Extending the canvas to a new aspect ratio
- D) Upscaling a low-resolution element to match the background's size

Q5

A generated patch of sky looks slightly softer and smoother than the rest of the photo. What fixes it?

- A) Regenerate until the sky comes out sharp
- B) Lower the generative layer's opacity
- C) Flatten the image so the patch merges
- D) Add matching grain and a touch of sharpening

Q6

A client composite used Generative Fill. What should you check before delivering?

- A) Which model produced it (Firefly or a partner model) and its terms for your account and deliverable
- B) Nothing: anything made in Photoshop is automatically licensed for any use
- C) Only that the file is under the client's size limit
- D) That the generative layers have been renamed

◆ JOURNAL

Where does generative fill genuinely save you time, and where does trusting it cost you more time cleaning up than doing it by hand? Draw your own line between the two. Also note: key concepts · what clicked · questions · ideas to try · how you feel.

MINDSET

New tools come and go, but the eye you have trained all course is what makes them useful. A weak variation is not a failure; it is one of three drafts. Curating is the real skill.

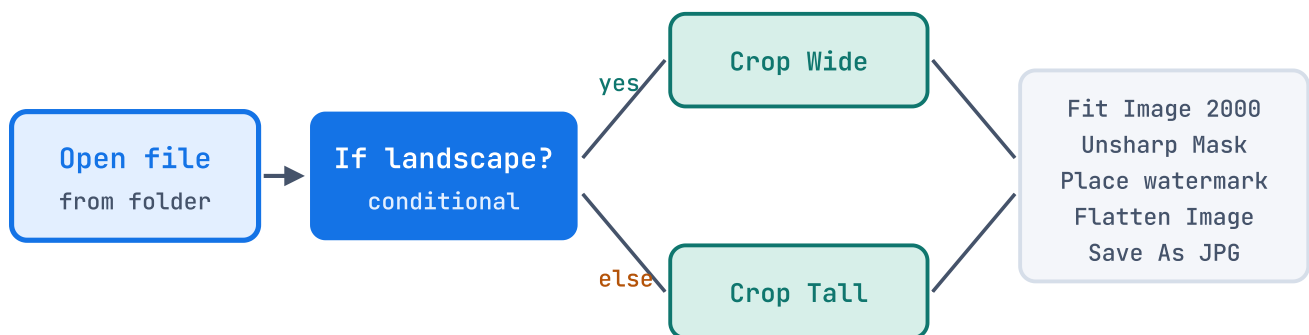
◆ NOTES

◆ BY THE END YOU CAN

- ◆ Record a clean, reusable **Action** and re-order or re-record steps
- ◆ Insert **stops, menu items, and modal pauses** for input
- ◆ Add a **Conditional** so one Action handles landscape and portrait differently
- ◆ **Batch** an Action across a folder and save out automatically
- ◆ Build a **droplet**, a drag-and-drop executable for your workflow

◆ KEY IDEAS

- ◆ **Action** = a recorded macro. **Conditional** = an *if* inside it. **Batch** = the Action pointed at a folder. **Droplet** = that batch saved as an icon you drop files onto.
- ◆ **Record relative, not absolute.** Image Size records fixed dimensions; File ► Automate ► **Fit Image** scales the long edge. Set Preferences ► Units & Rulers to **Percent** before recording crops and selections.
- ◆ **Actions capture what you did, not what you meant.** Record on a representative file; test on your weirdest one.
- ◆ **Insert Conditional** tests the document or active layer: Document Is Landscape, Portrait, or Square; Document Mode Is RGB or CMYK; Document Is 16 Bits Per Channel; Document Has Layers; Layer Is Type or Smart Object. **No** condition for pixel size or a layer's name.
- ◆ **A true batch asks no questions.** Override Action "Save As" Commands, Suppress File Open Options Dialogs, Suppress Color Profile Warnings, Log Errors To File, and dialog toggles off.



Crop first (branching by orientation), then Fit Image, so every output's long edge is 2000 px.

◆ WORKED EXAMPLE: A DELIVERY ACTION

Use a folder of mixed landscape and portrait **copies** (practice: `117-input/`), keep the watermark (`117-watermark.png`) outside that folder, and make an empty output folder beside it.

- 1 **Record the core.** Window ► Actions ► new set **Delivery**, new action **Web Delivery**, **Record**. File ► Automate ► **Fit Image** at 2000 × 2000 px (it scales the long edge to 2000), **Unsharp Mask**, **Place** your watermark PNG, **Layer** ► **Flatten Image**, **Save As** JPG. Stop recording.
- 2 **Add the branch.** Set Preferences ► Units & Rulers to **Percent**, then record two tiny helper actions in the same set: **Crop Wide** and **Crop Tall**. In Web Delivery, select the first step, then panel menu ► **Insert Conditional**: If Document Is Landscape play Crop Wide, else Crop Tall. The conditional lands **below** the selected step, so drag it to the top: crop first, then Fit Image, so the final long edge is still 2000 px.
- 3 **Batch it.** File ► Automate ► **Batch**: Action = Web Delivery, Source = your folder, Destination = an output folder, tick **Override Action "Save As" Commands**, set naming. Run, and watch landscape and portrait each crop correctly.
- 4 **Make a droplet.** File ► Automate ► **Create Droplet**, point it at Web Delivery, save it to your desktop. Drag a folder of new images onto it.

Record Layer ► Flatten Image before Save As: while the watermark is its own layer, Save As won't offer JPEG.

◆ YOUR TURN: THE PROJECT CHECKLIST

- ◆ ☐ A recorded, relative delivery Action (Fit Image, percent-based crops)
- ◆ ☐ A conditional branching landscape vs portrait
- ◆ ☐ A batch that runs the folder hands-off with Override Save As
- ◆ ☐ Errors logged, no mid-run dialogs
- ◆ ☐ A working droplet on the desktop

Write the steps your Actions panel shows for Web Delivery, in order:

The practice square file (03) is neither landscape nor portrait. Which branch did it take, and why?

◆ NOW YOU: SOLO VARIATION (START IN CLASS, FINISH AT HOME)

- ◆ ☐ An Action that outputs three sizes (thumb, web, print) into three subfolders in one run, batched with Destination set to **None**
- ◆ ☐ A conditional that only processes layered masters: If Document Has Layers, play the full delivery; else a lighter "already flat" version
- ◆ ☐ A droplet a non-Photoshop colleague could use, with a Stop step that explains what to do if something looks wrong

Tasks Actions couldn't handle (bring these to L18):

◆ WATCH OUT

- ◆ **Works on one image, breaks in the batch?** Almost always an absolute value: a fixed crop, a hard-coded Image Size, or a named layer that doesn't exist in every file. Re-record relative and route odd files with conditionals.
- ◆ **All files in one folder when you wanted subfolders?** Override Action "Save As" Commands sends every save to the Batch destination. Set Destination to **None** and let the Action's own Save As steps decide.
- ◆ **Rulers still in Percent?** Set them back to Pixels when you finish recording.

◆ ACTIONS VS DROPLETS VS SCRIPTS: WHICH ONE WHEN?

TOOL	BEST FOR
Action	Repeatable steps; with conditionals, repeatable steps on mixed input
Droplet	Handing that Action to someone else as a drag-and-drop icon (it runs when files are dropped on it; it doesn't watch a folder)
Script	Loops, math, external data, or decisions Actions can't express (scripts can even call Actions)

◆ RECIPE CARD: ACTIONS & BATCH

- 1 Window ► Actions → new set + action → Record relative steps (Fit Image; rulers in Percent)
- 2 Insert Conditional to branch (landscape / portrait / square / mode)
- 3 Insert stops / modals only where human input is truly needed
- 4 File ► Automate ► Batch over a folder, Override Save As, suppress dialogs, log errors
- 5 Create Droplet for a drag-and-drop reusable

Mantra: if you'll do it twice, record it once.

Q1

Why record File ► Automate ► Fit Image rather than Image ► Image Size in a delivery Action?

- A) Fit Image resamples with better quality
- B) Fit Image scales the long edge, so it works on landscape and portrait files alike
- C) Image Size can't be recorded in an Action
- D) Fit Image also sharpens the result

Q2

Before recording the "Crop Wide" and "Crop Tall" helper actions, what should you change so the crops replay relative to each image?

- A) Turn on the dialog toggle for the Crop step
- B) Record the crops on the largest file in the folder
- C) Set Preferences ► Units & Rulers to Percent
- D) Convert every file to 16 bits per channel

Q3

Which of these can Insert Conditional actually test?

- A) Whether the image is wider than 2000 pixels
- B) Whether a layer named "logo" exists
- C) Whether the file name contains "_raw"
- D) Document Is Portrait

Q4

You selected the first step of Web Delivery and inserted the conditional. What do you do next?

- A) Drag it to the top of the list so the crop runs before Fit Image
- B) Nothing; conditionals always run first
- C) Delete the Fit Image step, which the conditional replaces
- D) Move it to the end, after Save As

Q5

Which Batch settings make a run of 500 files hands-off?

- A) Destination: None and every dialog toggle turned on
- B) Errors: Stop For Errors, so you can fix each one
- C) Override Action "Save As" Commands, both Suppress options, and Log Errors To File
- D) Source: Opened Files, with all 500 files open

Q6

Your Action saves three sizes into three different subfolders. What should the Batch Destination be?

- A) Folder, with Override Action "Save As" Commands ticked
- B) None, and let the Action's own Save As steps decide
- C) Save and Close
- D) The same folder as the Source

◆ JOURNAL

List three repetitive tasks in your own workflow. For each, could an Action do it, or do you need a conditional or a script? Sorting them is the start of automating your practice. Also note: key concepts · what clicked · questions · ideas to try · how you feel.

MINDSET

An Action that breaks on its first batch is doing you a favor: it points at the one step that was too specific. Re-record that step, test on your weirdest file, and you have started thinking like a studio, not just an editor.

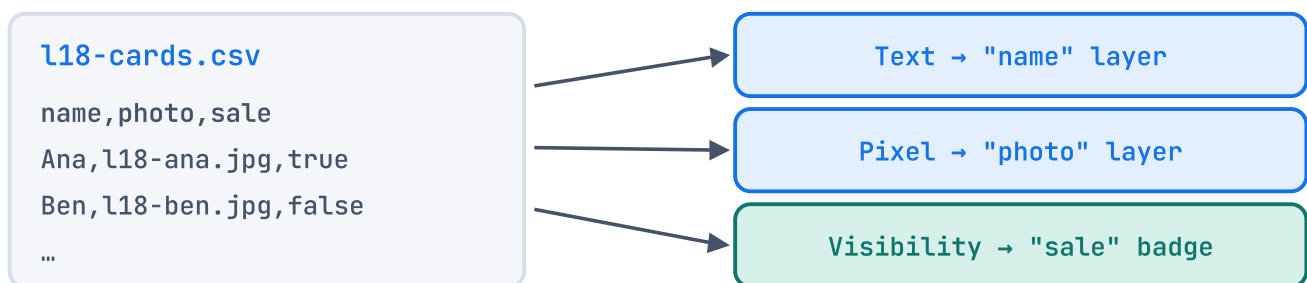
◆ NOTES

◆ BY THE END YOU CAN

- ◆ Define **Variables** that bind layers to data: text, visibility, and pixel replacement
- ◆ Import a **Data Set** from a CSV and preview each row on the canvas
- ◆ **Export Data Sets as Files** to render one graphic per row
- ◆ Understand where scripting begins and read a simple **ExtendScript** / JS example
- ◆ Choose between **Actions, Variables, and scripts** for a given job

◆ KEY IDEAS

- ◆ **Template + data.** The template is your design with certain layers marked as variable; the data is a table where each column maps to a variable and each row is one output.
- ◆ **Exactly three kinds of variable:** Text Replacement (swap the words in a type layer), Pixel Replacement (swap the image in a layer, with fit rules), Visibility (show or hide a layer).
- ◆ **The variable name binds.** Each variable binds to the CSV column whose header matches its name. Names can't contain spaces and must start with a letter, underscore, or colon. Naming the layer the same is a helpful habit, not a requirement.
- ◆ **The CSV rules:** UTF-8 with a header row; Visibility values `true` / `false` (Photoshop rejects yes/no); Pixel Replacement values as full file paths, or just file names with the images in the same folder as the CSV.
- ◆ **Export Data Sets as Files** renders one PSD per row (for JPG or PNG, run the PSDs through an Action).
- ◆ **Scripting begins** when you need loops, math, renaming by pattern, an API, or decisions based on pixel values. Photoshop runs ExtendScript (`.jsx`) and newer UXP scripts (`.psjs`) via File ► Scripts ► Browse.



Each CSV column drives one variable. Bind those three and most personalized-graphics briefs are solved.

◆ WORKED EXAMPLE: A DATA-DRIVEN BATCH

Design a simple card with a name (type layer), a photo (image layer), and a "SALE" badge (a layer to show/hide). Practice parts are in `l18-data/` : `l18-card-background.jpg` , six photos `l18-ana.jpg` ... `l18-fay.jpg` , `l18-sale-badge.png` , and `l18-cards.csv` . Save your template in that same folder. The photo layer must be a pixel layer (if you placed it as a Smart Object, Layer ► Rasterize ► Smart Object).

- 1 **Name everything.** Rename the layers exactly `name` , `photo` , `sale` . You'll give the variables the same names, and the CSV headers must match the variable names.
- 2 **Define variables.** Image ► Variables ► Define. On `name` tick **Text Replacement**; on `photo` tick **Pixel Replacement** (set the method: Fit, Fill, As Is, or Conform); on `sale` tick **Visibility**. Name each variable to match its CSV header.
- 3 **Import and proof.** Image ► Variables ► Data Sets... ► **Import** your CSV. Tick **Use First Column For Data Set Names**. With Preview on, step through a few rows and confirm the name, photo, and badge update on canvas. Fix any binding that didn't take (usually a header/variable-name mismatch, yes/no instead of true/false, or a photo path Photoshop can't find).
- 4 **Export the batch.** File ► Export ► **Data Sets as Files**. In File Naming, choose **Data Set Name**, add any fixed text you like, and export. Photoshop writes one PSD per row. Optional: run them through your L17 delivery Action to flatten to JPG.

◆ YOUR TURN: THE MODULE 6 DELIVERABLE CHECKLIST

- ◆ ☐ Variable names (and layers) matching the CSV headers exactly
- ◆ ☐ Text, Pixel, and Visibility variables defined
- ◆ ☐ CSV imported as a data set, several rows proofed on canvas
- ◆ ☐ Data Sets exported as files with a clear naming pattern
- ◆ ☐ (Optional) finished via a delivery Action / droplet

My three variables (layer · type · variable name · CSV header):

Anything that didn't bind the first time, and what fixed it:

◆ WHERE SCRIPTING BEGINS: READ IT BEFORE YOU WRITE IT

The lesson's example (practice copy: `l18-resize-folder.jsx`). Run it with File ► Scripts ► Browse, on a **copy** of a test folder: it writes its `web_` files into the folder it reads from.

```
// resize-folder.jsx - File ▶ Scripts ▶ Browse to run
var inputFolder = Folder.selectDialog("Choose a folder of images");
if (!inputFolder) { throw new Error("No folder chosen."); }
var files = inputFolder.GetFiles(/\. (jpg|jpeg|png)$/i);

for (var i = 0; i < files.length; i++) {
    var doc = app.open(files[i]);
    var longEdge = 2000;
    var scale = longEdge / Math.max(doc.width.value, doc.height.value);
    doc.resizeImage(doc.width * scale, doc.height * scale);

    var out = new File(inputFolder + "/web_" + doc.name.replace(/\.([^.] +$/ , ".jpg")
    var opt = new JPEGSaveOptions();
    opt.quality = 10;
    doc.saveAs(out, opt, true);
    doc.close(SaveOptions.DONOTSAVECHANGES);
}
alert("Done: " + files.length + " images resized.");
```

In plain English, what does the loop do to each file? Which two numbers would you change first?

◆ NOW YOU: SOLO VARIATION (START IN CLASS, FINISH AT HOME)

- ◆ ☐ A second visibility variable (a "VIP" ribbon) and a color swap via a visible/hidden tint layer, from new CSV columns (practice: `l18-cards-vip.csv` and `l18-vip-ribbon.png`)
- ◆ ☐ 20 social cards for a real list (a team roster, an event lineup), and a note of how long the manual version would have taken
- ◆ ☐ The sample script with a new long edge and output format, run on a test folder, plus a line that skips files already starting with `web_`

Manual time vs data-driven time for my 20 cards:

◆ ACTIONS, VARIABLES, OR A SCRIPT?

Same steps on every file (resize, sharpen, watermark, save)	Action + Batch
Same steps, but landscape and portrait need different handling	Action with a Conditional
Same layout, changing content (names, photos, show/hide)	Variables + Data Sets
Loops, math, renaming by pattern, external lookups, per-file logic	Script

◆ WATCH OUT

- ◆ **A stray space or capital letter** in a CSV header breaks the binding. Check the CSV in a text editor.
- ◆ **Pixel Replacement missing?** The photo layer is a Smart Object, type, or shape layer. It must be a pixel layer.
- ◆ **Proof before you export.** Step through a few rows on canvas before exporting five hundred.
- ◆ **Running the script?** Use a copy of the folder, and set Units & Rulers back to Pixels if you left them in Percent after L17.

◆ RECIPE CARD: DATA-DRIVEN

- 1 Name each variable to match its CSV header exactly (matching layer names is a helpful habit, not a requirement)
- 2 Image ► Variables ► Define: Text / Pixel / Visibility
- 3 Image ► Variables ► Data Sets... ► Import the CSV (first column as names; Visibility = true/false); proof rows on canvas
- 4 File ► Export ► Data Sets as Files, named by Data Set Name
- 5 Reach for scripting when you need loops, math, or logic

Mantra: design the difference as data, and the count stops mattering.

◆ SELF-CHECK

Q1

What are the three kinds of variable in Photoshop's Variables feature?

- A) Text, Color, and Size
- B) Layer Name, Layer Style, and Opacity
- C) Text, Smart Object, and Artboard
- D) Text Replacement, Pixel Replacement, and Visibility

Q2

What actually binds a variable to a column in the CSV?

- A) The layer's position in the Layers panel
- B) The variable name matching the CSV column header exactly
- C) The column's position, left to right
- D) The layer name, even if the variable is named differently

Q3

What values should the `sale` column use to show or hide the badge?

- A) `true` or `false`
- B) `yes` or `no`
- C) `show` or `hide`
- D) `1` or `0`

Q4

You want the **photo** column to list just file names like **118-ana.jpg** . What must be true?

- A) The images must be open in Photoshop during the import
- B) The images must be PSD files
- C) The images sit in the CSV's folder and the names match exactly (otherwise use full paths)
- D) The photo layer must be a Smart Object

Q5

You've proofed the rows. How do you render one file per row, named after each person?

- A) File ► Export ► Export As, with all layers selected
- B) File ► Automate ► Batch with the CSV as the Source
- C) Step through each row and use File ► Save As by hand
- D) File ► Export ► Data Sets as Files, naming by Data Set Name

Q6

Which job genuinely needs a script rather than Variables or an Action?

- A) 300 badges with the same layout and different names
- B) Resize every file in a folder, skipping any whose name already starts with **web_**
- C) Showing a SALE badge on some cards and not others
- D) Cropping landscape and portrait files differently

◆ JOURNAL

What in your work is really "one design, many values"? Sketch the template and the columns its data set would need. Seeing jobs as template-plus-data is the mindset shift of this module. Also note: key concepts · what clicked · questions · ideas to try · how you feel.

MINDSET

You don't need to be a programmer to adapt a snippet. You just read one and followed it. Start by tweaking values (the 2000, the quality); reading and adapting code is a real, employable skill on its own.

◆ NOTES

BEFORE SESSION 7

◆ FINISH MODULE 6

- ◆ ☐ My finished, layered L16 composite is saved (generative layers and masks intact)
- ◆ ☐ I noted one task where generation saved me time and one where I had to take over
- ◆ ☐ Web Delivery batched over a mixed folder with no mid-run dialogs; droplet tested on a fresh set
- ◆ ☐ **One data-driven batch shipped (10+ outputs) from a single template**
- ◆ ☐ The sample script adapted and run on a test folder
- ◆ ☐ Learning Journal entries written for L16, L17, and L18

COMING UP: SESSION 7

Module 7, Multi-Shot Techniques: panoramas, HDR, focus stacking, and timeline animation. Bring any overlapping, bracketed, or focus-stack frames you have (practice sets are provided). Session 7 also kicks off your capstone with Lesson 8.1, the brief and the plan, so start thinking about what you want to make.

YOU FINISHED MODULE 6 🎉

You added speed and scale to your craft: AI assistance for the tedious parts, recorded and conditional automation, and data-driven production plus a first real taste of scripting. That is the difference between an artist and a one-person studio.

Multi-Shot Techniques

Welcome to the Going Further tier. A single frame is only so wide, holds only so much range, and has only so much in focus. Today you beat all three limits by merging several frames into one, then you turn layers into motion. The session ends with the capstone kickoff: you leave with a brief and a plan.

Lessons 19–20 (7.1–7.2)

2 lessons + capstone kickoff • Session 7

Name: _____

HOW TO USE THIS WORKBOOK

Each lesson has **By the end you can**, **Key ideas**, a **Worked example** (the guided build from the lesson), a **Your turn** checklist with space to write, a **Recipe card** to keep, a **Self-check** quiz, and a **Journal** prompt. Journal after every lesson: key concepts · what clicked · questions · ideas to try · your progress and feelings.

PRACTICE FILES FIRST, YOUR OWN SETS AT HOME

Merges are heavy work for a computer. In class, use the practice sets your instructor gives you (they merge in seconds). Save your own full-resolution shoots for the "Now You" at home, when a long progress bar costs nobody else any time.

MODULE PROGRESS TRACKER

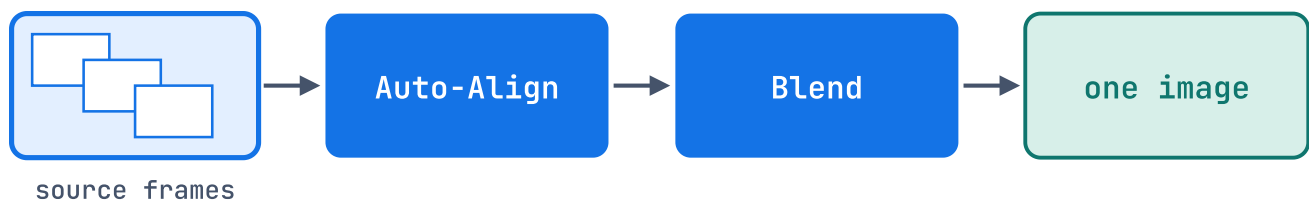
- ◆ ☐ **L19:** I stitched a panorama, merged an HDR bracket, and focus-stacked a series, and checked every join at 100%
- ◆ ☐ **L20:** I keyframed a seamless loop and exported it as a GIF and an MP4
- ◆ ☐ **Kickoff:** I wrote my capstone brief, started my moodboard, and drafted my plan (on the Module 8 workbook's L21 pages)
- ◆ ☐ **Homework:** plan finished, capstone build well underway before Session 8

◆ BY THE END YOU CAN

- ◆ Shoot overlapping frames and stitch a panorama with **Photomerge**
- ◆ Bracket exposures and combine them with **Merge to HDR Pro** for wider range
- ◆ Shoot a focus series and focus-stack it with **Auto-Align + Auto-Blend**
- ◆ See the shared engine (auto-align, then blend) behind all three
- ◆ Clean the seams, ghosts, and edges these merges leave behind

◆ KEY IDEAS

- ◆ **Align, then blend.** Auto-Align finds common detail and warps the frames into registration. Blend then chooses, per pixel, the best contribution: the overlap for a panorama, the well-exposed tone for HDR, the sharpest region for a stack.
- ◆ **Change one variable only.** Angle for a panorama, shutter (exposure) for HDR, focus distance for a stack. Hold everything else as steady as you can; a tripod and a 2-second timer help most for HDR and stacking.
- ◆ **Three front doors:** File ► Automate ► **Photomerge**; File ► Automate ► **Merge to HDR Pro**; File ► Scripts ► **Load Files into Stack**, then Edit ► **Auto-Align Layers** and Edit ► **Auto-Blend Layers...** (Stack Images).
- ◆ **HDR output:** 16 Bit (or 8 Bit) tone-maps right in the dialog; **32 Bit** keeps the full range to tone-map later with Camera Raw.
- ◆ **Garbage frames in, garbage merge out.** Too little overlap won't stitch; movement mid-bracket leaves ghosts; a skipped focus distance leaves a soft band.



pano: vary angle → wider field HDR: vary exposure → range stack: vary focus → depth

One engine, three jobs. Whatever you varied between frames, Photoshop aligns first, then blends the best of each.

◆ WORKED EXAMPLE: MERGE THREE WAYS

- 1 **Panorama.** File ► Automate ► Photomerge, Browse to the pan frames, Layout **Auto**, tick **Blend Images Together** and **Content Aware Fill Transparent Areas**, OK. Crop or accept the filled edges; heal any obvious stitch artifact.
- 2 **HDR.** File ► Automate ► Merge to HDR Pro, add the bracket, tick **Remove Ghosts** if anything moved. For a finished file now, set Mode to **16 Bit** and tone-map in the dialog (keep it believable). To keep every stop, set **32 Bit** and tone-map later with Camera Raw.
- 3 **Focus stack.** File ► Scripts ► Load Files into Stack (tick **Attempt to Automatically Align Source Images**). Select all layers, Edit ► Auto-Blend Layers..., choose **Stack Images**, tick **Seamless Tones and Colors**. Mask-fix any soft patch.
- 4 **Clean the joins.** Zoom to 100% on each result and heal seams, ghosts, or soft bands. Apply a Module 4 unifying grade so all three feel like your work.

Shooting your own: overlap pan frames by roughly 30% (Photomerge doesn't measure it for you); bracket -2 / 0 / +2; step focus in small, even moves from front to back.

◆ YOUR TURN: MERGE THREE WAYS

Use the practice sets (or your own small sets). Tick each item as you finish it:

- ◆ ☐ A stitched panorama with invisible seams
- ◆ ☐ An HDR merge with recovered highlights and shadows, no ghosts
- ◆ ☐ A focus stack sharp front-to-back
- ◆ ☐ Edges cropped/filled, joins healed at 100%
- ◆ ☐ A consistent grade across all three

What did you find at 100% on each result, and how did you fix it? (pano · HDR · stack)

My HDR settings: Mode _____ · Method _____ · Remove Ghosts ☐ · What I pulled back to keep it believable:

◆ WATCH OUT

- ◆ **Panorama bows or bends?** Try Cylindrical or Spherical in Photomerge's Layout list, keep the horizon centered when you shoot, and straighten with Filter ► Adaptive Wide Angle.
- ◆ **Auto-Blend grayed out?** Select *all* the layers first. And choose Stack Images, not Panorama.
- ◆ **Photoshop asks you to "Manually Set EV"?** Your files lost their exposure data. Enter the values from your instructor's lab sheet (or your camera's settings for your own bracket).

◆ RECIPE CARD: MULTI-SHOT MERGE

- 1 Panorama: Photomerge (Auto + Blend + Content-Aware edges)
- 2 HDR: Merge to HDR Pro (bracket, Remove Ghosts, 16-bit tone-map or 32-bit to keep range)
- 3 Focus stack: Load Files into Stack → Auto-Align → Auto-Blend Layers... (Stack Images)
- 4 Overlap generously; change only the one intended variable
- 5 Heal seams/ghosts at 100%, then one shared grade

Mantra: align first, blend second, and one careful shoot beats any repair.

◆ SELF-CHECK

Q1

What do panoramas, HDR, and focus stacking have in common?

- A) Each one needs a 32-bit file to work
- B) Photoshop auto-aligns the frames, then blends the best of each
- C) Each uses Generative Fill to invent the missing detail
- D) Each starts from one frame duplicated several times

Q2

You are shooting a focus stack. What should change from one frame to the next?

- A) The camera angle
- B) The shutter speed
- C) The focus distance
- D) The white balance

Q3

Your Photomerge result has ragged, transparent edges. Which option would have filled them for you?

- A) Vignette Removal
- B) Geometric Distortion Correction
- C) Layout: Reposition
- D) Content Aware Fill Transparent Areas

Q4

A bird flew across the window between your three bracketed shots, and the merge shows three faint birds. What do you do?

- A) Tick Remove Ghosts in Merge to HDR Pro
- B) Switch the Mode to 8 Bit
- C) Merge the bracket with Photomerge instead
- D) Raise the Detail slider

Q5

You want to keep every stop of range now and do the tone-mapping later. Which Merge to HDR Pro setting fits?

- A) 8 Bit, tone-mapped in the dialog
- B) 32 Bit, then tone-map later in Camera Raw
- C) 16 Bit with Local Adaptation
- D) Any mode; HDR Pro always keeps the full range

Q6

Your focus stack is sharp in front and at the back, but a soft band runs across the middle. What is the most likely cause and fix?

- A) Seamless Tones and Colors was ticked; untick it and blend again
- B) The layers were not Smart Objects; convert them first
- C) Auto-Align overcorrected; turn it off
- D) A focus distance was skipped, so reshoot with smaller steps or mask in a sharper frame

◆ JOURNAL

Which single-frame limit frustrates you most: width, range, or depth? Plan the shoot that would beat it with a multi-shot merge. Also note: key concepts · what clicked · questions · ideas to try · how you feel.

MINDSET

Multi-shot work rewards patience at the camera more than skill at the keyboard. If a merge fails, it is usually telling you something about the frames, not about you: more overlap, a steadier hand, a smaller focus step. That is information for the next shoot.

◆ NOTES

- ◆ Choose between **Frame Animation** and the **Video Timeline**
- ◆ Build a frame loop with tweening
- ◆ Animate layer position, opacity, and style with keyframes (and scale, via Smart Objects)
- ◆ Loop cleanly and shape timing so motion feels natural
- ◆ Export a GIF (Save for Web) or an MP4 (Render Video)

◆ KEY IDEAS

- ◆ **Frame Animation** is the flip-book: discrete frames, each a layer arrangement with its own hold time; **Tween** generates the in-betweens. Best for short, loopy GIFs with exact timing.
- ◆ **Video Timeline** is the smooth path: each layer gets a track; click a property's **stopwatch** to drop a keyframe, move the playhead, change the value, and Photoshop adds the next keyframe and interpolates between them.
- ◆ **Tracks depend on the layer.** Normal layers get Position, Opacity, and Style. Smart Objects get a **Transform** track instead of Position, which is how you animate scale and rotation. Warps can't be keyframed.
- ◆ **No built-in ease.** Keyframes interpolate linearly (or Hold, for a sudden jump). Fake an ease with an extra keyframe near the end of a move.
- ◆ **Two doors out:** File ► Export ► **Save for Web (Legacy)** for a GIF (256 colors max, heavy); File ► Export ► **Render Video** for an H.264 MP4 (smaller, full color).

0.1s 0.1s 0.1s 1.0s

discrete cells • exact timing

A diagram showing a light blue rounded rectangle containing a dashed line. At each end of the dashed line is a yellow diamond. Below the left diamond is the word "start" and below the right diamond is the word "end".

keyframes · Photoshop fills the middle

Frames give you flip-book control; keyframes give you smooth motion the app fills in between the dots.

WORKED EXAMPLE: A LOOPING ANIMATION

- 1 Open a simple layered design (a logo, a badge, a headline). Convert anything you'll scale to a Smart Object.
- 2 **Start the timeline.** Window ► Timeline, click **Create Video Timeline**. Panel menu ► Set Timeline Frame Rate ► **24 fps**. Set a 2–3 s duration: drag the right end of each layer's bar to the 3-second mark, then drag the work-area end handle (above the ruler) to match.
- 3 **Keyframe the moves.** On the badge (a Smart Object), twirl open and click the **Transform** stopwatch at frame 0 with the badge small. Move the playhead, scale it to its final size and spot; a keyframe drops automatically. On the headline, keyframe **Opacity** 0 → 100 and a small upward **Position** move so it rises in.
- 4 **Loop and shape the timing.** Hold the finished pose for a beat, then add keyframes that return each layer to its frame-0 state by the last frame. To fake an ease, add an extra keyframe close to the end of each move with only a small change left. Play back and adjust until it feels natural.
- 5 **Export both.** Save for Web (Legacy) (Ctrl+Alt+Shift+S / ⌘+⇧+S) → GIF, Looping **Forever**, tuned palette. Render Video → **H.264**. One animation, two delivery formats.

◆ YOUR TURN: A LOOPING ANIMATION

- ◆ ☐ A video timeline with keyframed Transform (or Position) and Opacity
- ◆ ☐ Transformed layers converted to Smart Objects (stay sharp)
- ◆ ☐ A seamless loop with natural-feeling timing (extra keyframes for a slow finish)
- ◆ ☐ Exported a GIF (looping forever) and an MP4
- ◆ ☐ GIF kept short and reasonably small

My keyframe plan. For each layer: which property, and what happens at the start, the hold, and the end?

File sizes: GIF _____ · MP4 _____. What did I change to make the GIF smaller?

◆ WATCH OUT

- ◆ **No scale track?** It's a normal layer. Convert to Smart Object and the Transform track appears.
- ◆ **Loop jumps at the wrap-around?** The last frame doesn't match the first, or the work area runs past your last keyframes.
- ◆ **GIF banded or huge?** GIF is limited to 256 colors and grows with size, colors, and frames. Crop tight, keep it a few seconds, or deliver an MP4.

◆ RECIPE CARD: ANIMATE & EXPORT

- 1 Smart-Object anything you'll transform; Window ► Timeline
- 2 Video timeline → keyframe Position / Opacity / Style, or Transform on Smart Objects (stopwatch)
- 3 Return to the frame-0 pose by the last frame; fake easing with extra keyframes
- 4 Save for Web → GIF (loop Forever): small & short
- 5 Render Video → H.264 MP4 for full-color playback

Mantra: keyframe the start and end, let Photoshop travel the middle.

◆ SELF-CHECK

Q1

You want a smooth glide and fade, and you plan to export an MP4. Which mode do you start in?

- A) Frame Animation
- B) The Video Timeline
- C) Artboards
- D) An Action recorded with playback pauses

Q2

The badge must grow from small to full size, but its track only shows Position, Opacity, and Style. What do you do?

- A) Keyframe Position instead; it scales the layer too
- B) Keyframe a Style change
- C) Convert the badge to a Smart Object so it gets a Transform track
- D) Rasterize the layer first

Q3

What makes a loop seamless?

- A) Returning each layer to its frame-0 state by the last frame
- B) Setting the frame rate to 30 fps
- C) Adding a Hold keyframe on frame 0
- D) Setting Looping Options to Once

Q4

Your motion feels robotic. Photoshop keyframes move at a constant speed. How do you fake an ease?

- A) Choose Ease In from the keyframe's right-click menu
- B) Switch every keyframe to Hold interpolation
- C) Raise the frame rate to 60 fps
- D) Add an extra keyframe near the end of each move with only a small change left

Q5

Where do you export a looping GIF?

- A) File ► Export ► Export As, GIF
- B) File ► Export ► Save for Web (Legacy), GIF, Looping Options: Forever
- C) File ► Export ► Render Video, GIF preset
- D) File ► Save As, GIF

Q6

Your gradient-heavy loop bands badly as a GIF and weighs 4 MB. What is the better deliverable?

- A) The same GIF with the colors cut to 16
- B) A single PNG of the last frame
- C) An MP4 made with Render Video (H.264)
- D) The GIF exported at 200% size

◆ JOURNAL

What is one still design of yours that a few seconds of motion would improve, and which single element should move? Restraint matters in animation too. Also note: key concepts · what clicked · questions · ideas to try · how you feel.

MINDSET

Your first loop will probably feel stiff. That is normal: timing is learned by playing it back, nudging one keyframe, and playing it again. A few small moves done well beat a busy animation every time.

◆ NOTES

CAPSTONE KICKOFF: THE BRIEF & THE PLAN

TONIGHT YOU START LESSON 8.1 (L21)

The last part of the session runs Lesson 8.1 in class. You work on the **L21 pages of the Module 8 workbook** (your instructor hands them out): a one-page brief with six parts (objective, audience, deliverables + specs, constraints, success criteria, references), a moodboard, and a staged plan. Bring those pages back to Session 8, where you review your plan with a partner before the studio block.

◆ QUICK CAPTURE: MY CAPSTONE IN ONE SENTENCE

Write your objective in one sentence. If you can't say what success looks like in a sentence, the idea isn't ready yet.

BEFORE SESSION 8: HOMEWORK

PLAN ON BUILDING AT HOME

Session 8 opens with a plan review in pairs, then a studio block for **finishing**, then presentations and critique. It is not enough time to build a capstone from scratch, so arrive with your piece well underway.

◆ PART 1: FINISH THE PLAN (LESSON 8.1)

- ◆ ☐ My one-page brief has all six parts, and someone else has read my success criteria
- ◆ ☐ Every vague word is pinned to a spec or a reference
- ◆ ☐ My moodboard has 6–12 references, each annotated with the one thing I'm taking from it
- ◆ ☐ My plan maps each stage to a module skill, lists my assets, and has 2–3 milestones; stretch goals are marked optional
- ◆ ☐ My source assets are gathered, and I've checked the license on anything I didn't shoot
- ◆ ☐ L21 journal: my one-sentence objective

◆ PART 2: BUILD IT (LESSON 8.2)

- ◆ ☐ Worked in passes: block-in, refine, polish
- ◆ ☐ Groups and layers named as I went; Smart Objects and adjustment layers kept live
- ◆ ☐ Versioned saves so I can always step back
- ◆ ☐ I wrote down the one or two things I most want help with in the studio block

◆ MODULE 7 WRAP-UP

- ◆ ☐ One multi-shot still of my own (pano, HDR, or stack)
- ◆ ☐ One short animation, exported as a GIF and an MP4
- ◆ ☐ L19 and L20 journal entries written

YOU FINISHED MODULE 7 🎉

You went wider, higher-range, deeper, and into motion, further than most Photoshop users ever go. Only the capstone remains: the project that turns everything you have learned into something you can show the world.

Capstone

No new tools, only you and everything you have learned. One self-directed professional project, taken from a one-page brief to a presented case study: brief it, build it in passes, then tell its story.

Lessons 8.1–8.3 (L21–L23)

Kickoff in Session 7 • finale Session 8

Name: _____

MOST OF THE CAPSTONE HAPPENS AT HOME

You start the brief in class at the end of Session 7. Finish the brief, moodboard, and plan at home, then build. **Arrive at Session 8 with Pass 1 and Pass 2 done and Pass 3 under way**, about 70 to 80% of the piece. In class, the studio block is for finishing, fixing, and getting help, and half the session is case studies and presentations. **Screenshot the end of every pass** and keep a true "before": the case study needs them.

HOW TO USE THIS WORKBOOK

L21 is your one-page brief and plan: fill it in before you open Photoshop. L22 is your studio log and self-critique. L23 holds your case-study story, your 30-second version, and the feedback you receive. Your capstone is scored with the same rows you use to judge yourself here. Each lesson also has a Self-check and a Journal prompt: key concepts • what clicked • questions • ideas to try • your progress and feelings.

CAPSTONE TRACKER

- ◆ ☐ **Session 7 kickoff:** brief started, project chosen, moodboard file created
- ◆ ☐ **L21 (home):** brief finished, success criteria read by someone else, 6–12 annotated references, staged plan with milestones
- ◆ ☐ **L22 (home):** Pass 1 and Pass 2 done, Pass 3 under way; versions saved; end-of-pass screenshots
- ◆ ☐ **Session 8:** finished, success criteria ticked, deliverables exported at spec, presented
- ◆ ☐ **L23 (within a week):** case study laid out, exported, and published

◆ BY THE END YOU CAN

- ◆ Write a tight creative brief: objective, audience, deliverables, constraints, success criteria
- ◆ Translate a vague idea into **specific, checkable** requirements
- ◆ Build a moodboard that fixes the visual direction before you start
- ◆ Draft a staged plan that maps each phase to the skills you have learned
- ◆ Scope realistically so the project is ambitious but finishable

◆ KEY IDEAS

- ◆ **The brief is the contract** between the vision and the work, with a client or with your future self at 2 a.m. wondering if you are done. Its job is to make success **checkable**.
- ◆ **Specificity is the whole game.** Pin every vague adjective to a reference image, a measurable spec, or a named constraint.
- ◆ **Idea → brief → plan.** Each step removes ambiguity, so by the time you open Photoshop there are no unanswered questions to stall you.
- ◆ **A moodboard** of 6–12 annotated references fixes the visual target while it is still cheap to change, and becomes the yardstick you grade drafts against.
- ◆ **Scope for finishing, not for showing off.** A tightly executed simple idea beats an ambitious half-finished one. Stretch goals must be cuttable.

◆ WORKED EXAMPLE: FROM STICKY NOTES TO A SPEC

VAGUE NOTE	PINNED TO...
"Make it modern"	"A 3000×2000 hero image for a coffee brand's homepage, warm and editorial, readable at a glance on mobile, delivered as web + print by Friday"
"Soon-ish"	A date and a time (a constraint)
"Our colors, I guess?"	Two hex values (a constraint)
"Make it pop"	"Product name readable on a phone-sized thumbnail" (a success criterion)
"Clean but bold"	Two annotated images on the moodboard (a reference)

Your turn in class: **l21-vague-brief.txt** . Could a stranger build against your version and know when it is finished?

◆ PICK YOUR PROJECT

PROJECT	WHAT IT IS	TOUCHES
☐ Key art poster	A composited hero with title treatment	Modules 1, 3, 4, 5
☐ Product campaign	A hero plus a sized social set from one master	Modules 2, 4, 5, 6
☐ Editorial portrait	A retouched, graded portrait with layout and type	Modules 2, 3, 5
☐ Your own	Anything that lets you show the skills you most want on your portfolio	You choose

No client? Invent one: a fictional brand, a favorite band's next album, a film that doesn't exist. Stuck? Ask for **l21-backup-briefs.txt** .

◆ STEP 1: WRITE THE BRIEF (15 MIN) • ONE PAGE, SIX PARTS

1. Objective: the one sentence on what this must achieve and why.

2. Audience: who it is for; their context and what should make them stop.

3. Deliverables + specs: exact outputs, sizes, formats, and where they will live.

_____ W _____ H _____ ppi _____ color _____ format _____
_____ W _____ H _____ ppi _____ color _____ format _____

4. Constraints: brand rules, colors, fonts, budget, and the deadline.

5. Success criteria: a checklist you could hand to someone else to grade.

☐ _____

- ☐ _____
- ☐ _____
- ☐ _____
- ☐ _____

6. References / moodboard: see Step 2.

◆ STEP 2: BUILD THE MOODBOARD (15 MIN)

Collect 6–12 references for tone, color, type, and composition. A contact-sheet PSD or a simple grid is fine. Annotate each with **the one thing you are taking from it**.

#	REFERENCE (WHAT / WHERE FROM)	THE ONE THING I'M TAKING FROM IT
1		
2		
3		
4		
5		
6		
7–12		

◆ STEP 3: DRAFT THE PLAN (10 MIN)

List the stages and the module skill each uses. Cross out the stages your project does not need. Note the assets you must source or shoot (your own photos, or stock whose license and credit terms you have checked).

STAGE	MODULE SKILL	ASSETS TO SOURCE OR SHOOT
Assets & cutouts	Module 1 · masking	
Color base	Module 2 · grade/space	
Subject retouch	Module 3	
Composite	Module 4 · light/shadow/atmos	
Type & system	Module 5	
Automate & export	Module 6 · maybe 7	
Present	Module 8.3	

Milestones (two or three) and a realistic finish date:

1. _____ by _____ 2. _____ by _____ 3.
_____ by _____

Stretch goals (optional; each must be cuttable without breaking the deliverable):

◆ PROJECT COMPLETION CHECKLIST

- ◆ ☐ A one-page brief with all six parts
- ◆ ☐ Every vague term pinned to a spec or reference
- ◆ ☐ Success criteria written as a checkable list
- ◆ ☐ An annotated moodboard of 6–12 references
- ◆ ☐ A staged plan mapping skills, assets, and milestones

Session 8 plan review. My partner tried to grade my file against my success criteria. Criteria they could not check (rewrite them):

My studio goal for Session 8 (the one or two things I will finish, and what I need help with):

◆ RECIPE CARD: THE BRIEF

- 1 Six parts: objective · audience · deliverables+specs · constraints · success · references
- 2 Pin every adjective to a spec or a moodboard image
- 3 Write success criteria as a checkable list
- 4 Stage the plan; map each stage to a module skill
- 5 Scope to finish; make stretch goals cuttable

Mantra: if you can't check it, it isn't in the brief.

◆ SELF-CHECK

Q1

Which list is the six parts of a one-page brief?

- A) Title, headline, body copy, image, logo, call to action
- B) Tools, filters, fonts, layers, actions, export settings
- C) Objective, audience, deliverables + specs, constraints, success criteria, references/moodboard
- D) Budget, deadline, client name, file name, color mode, resolution

Q2

A client writes "make it modern." What does the lesson say to do with it?

- A) Pin it to something concrete: a reference image, a measurable spec, or a named constraint
- B) Leave it in; the client will recognize modern when they see it
- C) Replace it with a synonym such as "contemporary"
- D) Delete it; adjectives never belong in a brief

Q3

How should the moodboard be built?

- A) One perfect reference that you copy closely
- B) As many images as you can find; the more the better
- C) Screenshots of the Photoshop techniques you plan to use
- D) 6–12 references for tone, color, type, and composition, each annotated with the one thing you are taking from it

Q4

Which success criterion is written the way the lesson asks?

- A) "It should feel premium."
- B) "The product name is readable on a phone-sized thumbnail."
- C) "It looks as good as the references."
- D) "The client will love it."

Q5

What makes a stretch goal safe to include in your plan?

- A) It can be cut without breaking the deliverable
- B) It is the most impressive part of the piece
- C) It uses a skill you have never tried
- D) It is scheduled first, while your energy is high

Q6

According to the lesson, what is the commonest capstone failure?

- A) Choosing a simple idea
- B) Spending too long on the moodboard
- C) Writing the brief before opening Photoshop
- D) Over-scoping: an ambitious idea that ends half-finished

◆ JOURNAL

Write your capstone's one-sentence objective. If you can't say what success looks like in a sentence, the idea isn't ready; keep refining until you can. Also note: key concepts · what clicked · questions · ideas to try · how you feel.

MINDSET

Planning can feel like a delay, but it's the part that makes finishing possible. A project with a clear finish line is more than most ambitious ideas ever get. The hard thinking is done; the fun part starts next.

◆ BY THE END YOU CAN

- ◆ Execute a project in passes: **block-in, refine, polish**
- ◆ Keep ruthless layer hygiene: **groups, names, color labels**
- ◆ Stay non-destructive with Smart Objects and adjustment layers throughout
- ◆ Version and save so you can always step back safely
- ◆ Use fresh-eyes tricks to judge your own work honestly

◆ KEY IDEAS

- ◆ **Work in passes.** Raise the whole image a level at a time. Each pass is done across the frame before you start the next, never one perfect corner at a time. Zoom out between passes and judge the frame, not the pixel.
- ◆ **Layer hygiene.** Group by role, top-to-bottom in render order, name everything, color-label groups. Every adjustment a live layer, every scaled element a Smart Object, every filter a Smart Filter.
- ◆ **Save versions, and save often:** `capstone_v01`, `v02` at each pass, plus a flattened "reference" JPG of milestones.
- ◆ **Done** is when a pass changes almost nothing and every brief success criterion is met, not when you run out of things you could do.

◆ WORKED EXAMPLE: AN ORGANIZED STACK

GROUP (TOP TO BOTTOM)	WHAT LIVES THERE
▶ ★ Grade	Curves (and a LUT): the single unifying grade on top
▶ Type & layout	The type system (Module 5.1) and templated elements (5.2)
▶ Subject	A Smart Object, cut out (Module 1) and retouched (Module 3)
▶ Atmosphere & light	Shadows, reflections, haze, rim light (Module 4)
▶ Background	The plate, placed at the right scale for the horizon

Groups by role, in render order, each color-labeled: a stranger (or future you) can navigate this file in seconds.

◆ YOUR TURN: THE STUDIO SESSION, PASS BY PASS

Open a new document at your brief's largest size. Bring in your plan and moodboard. Tick each pass when it is done **across the whole frame**, save a version, and take the screenshot.

PASS	WHAT	DONE · VERSION · SCREENSHOT
1 · Block-in	Source and roughly place every element; cut out subjects (M1); scale and position against the horizon (4.1); set up and name groups now, while the file is small	☐ v____ ☐
2 · Refine	Correct color and set the working space (M2); retouch subjects (M3); match light and build shadows (4.2); tighten every mask; lay in the type system (5.1) and templated elements (5.2)	☐ v____ ☐
3 · Polish	Depth and atmosphere, then the single unifying grade on top (4.3); multi-shot elements if your brief uses them (M7); rim light, grain, vignette; automate any repetitive export (M6)	☐ v____ ☐
Finish · the gate	Open your 8.1 success criteria and tick each one on the artwork; save the layered master and export the deliverables at spec	☐ master ☐ exports

Studio log (date · pass · what I did · version saved · what's next):

◆ JUDGING YOUR OWN WORK (SCORE YOURSELF BEFORE CALLING IT DONE)

These are rows 2–5 of the capstone rubric. Run each check, then tick.

CHECK	HOW	STRONG / NOT YET
2 · Meets the brief	Open the success criteria and tick each one on the artwork. Grade against the brief, not your mood.	☐ ☐
3 · Composition & balance	View ► Flip Horizontal (it flips only the view; older versions: Image ► Image Rotation ► Flip Canvas Horizontal, so flip back after). Errors leap out in the mirror.	☐ ☐
4 · Focal point & read	Zoom to fit and squint. Does the focal point and read survive at thumbnail size and blurred?	☐ ☐
5 · Tonal structure	A temporary Black & White adjustment layer at the very top of the stack (toggle it off when done). Any tonal problems the color was hiding?	☐ ☐

Walk away: even ten minutes resets your eye; a night is better. What did you notice when you came back?

◆ PROJECT COMPLETION CHECKLIST

- ◆ ☐ Built in passes, whole-image at each level
- ◆ ☐ Grouped, named, color-labeled, non-destructive layer stack
- ◆ ☐ Incremental versions saved along the way
- ◆ ☐ Fresh-eyes checks (flip, squint, desaturate, walk away) done
- ◆ ☐ Every brief success criterion met and ticked
- ◆ ☐ Layered master saved; deliverables exported at spec

◆ WATCH OUT

- ◆ **One perfect corner.** Hours at 300% on a detail you may later cut. Finish the pass across the frame first.
- ◆ **File slow and huge?** Save a full layered version first. Then: convert heavy element groups to Smart Objects (or linked ones), rasterize truly final elements you're sure of, Edit ► Purge ► Histories, and keep working resolution sensible. Good hygiene from Pass 1 is the real fix.
- ◆ **Check layer left on.** Toggle the Black & White layer off before you export.
- ◆ **Moving things a pixel back and forth?** You're past done: save it and present it.

◆ PUSH IT FURTHER (OPTIONAL)

- ◆ An alternate grade or a day/night variant from the same master (Modules 4.3, 5.2)
- ◆ Spin the hero into a sized campaign set (Module 5.3) or a short animated version (Module 7.2)
- ◆ Record the repetitive export as an Action or data set (Module 6)

◆ RECIPE CARD: STUDIO SESSION

- 1 Pass 1 block-in (place, scale, group + name early)
- 2 Pass 2 refine (color, retouch, light/shadow, masks, type)
- 3 Pass 3 polish (atmosphere, unifying grade, details)
- 4 Stay non-destructive; save versions each pass
- 5 Fresh-eyes check, then grade against the brief

Mantra: raise the whole image a level at a time, and keep every decision reversible.

Q1

What does "work in passes" mean?

- A) Finish one corner perfectly, then move to the next
- B) Raise the whole image a level at a time: block-in, then refine, then polish, each done across the frame
- C) Save a new file after every single edit
- D) Do all the type first, then all the imagery

Q2

How does the lesson say to protect a bold experiment mid-project?

- A) Rely on History snapshots; they are saved with the file
- B) Duplicate the Background layer before each pass
- C) Flatten the file first so there is less to lose
- D) Save incrementally (capstone_v01, v02) at each pass and keep a flattened reference JPG of milestones

Q3

Which layer stack keeps every decision reversible to the end?

- A) Adjustments as live layers, scaled elements as Smart Objects, filters as Smart Filters, grouped by role and color-labeled
- B) Everything merged into one layer per pass to keep it fast
- C) Filters applied directly to a duplicate of the photo layer
- D) Layers named at the end, once the image is finished

Q4

What does View ► Flip Horizontal do, and why use it?

- A) Mirrors the pixels of the active layer to fix the composition
- B) Creates a mirrored copy of the document for export
- C) Flips only the view, so composition and balance errors leap out in the mirror
- D) Flips the canvas to add print bleed

Q5

How do you run the desaturate check without harming the file?

- A) Image ► Mode ► Grayscale, then undo
- B) A temporary Black & White adjustment layer at the very top of the stack, toggled off when done
- C) Drag every layer's Saturation to -100
- D) Export a grayscale JPG and compare

Q6

When is the capstone artwork actually finished?

- A) When you run out of things you could do
- B) When you are too tired to continue
- C) When the file passes 100 layers
- D) When a fresh-eyes pass changes almost nothing and every brief success criterion is met

◆ JOURNAL

Which pass was hardest to leave "good enough" and move on from? Learning when to stop refining a corner is a professional skill worth naming. Also note: key concepts · what clicked · questions · ideas to try · how you feel.

MINDSET

Finishing is its own skill. Every pass you push through and every "good enough, move on" call you make is the judgment that separates working professionals from perpetual tinkers.

◆ BY THE END YOU CAN

- ◆ Structure a case study: **challenge → approach → process → result → reflection**
- ◆ Present the final piece **in context** with mockups, not floating in space
- ◆ Use before/after and progress shots to show your thinking
- ◆ Write about your work clearly, for clients and employers
- ◆ Package and export a portfolio-ready presentation

◆ KEY IDEAS

- ◆ **Pretty pictures** say "I can operate Photoshop." **Case studies** say "I can solve problems." Show the thinking, not just the pixels.
- ◆ **Show the decisions, not every step.** A highlight reel, not a tutorial: the three or four moments that show judgment.
- ◆ **Context** tells the reviewer what the thing is for and makes your work look shipped rather than sketched. Use the Smart-Object mockups from Module 5.2.
- ◆ **Keep before/after honest.** A fabricated "before" is obvious to anyone who does the work. Disclose any generative assistance.

◆ WORKED EXAMPLE: THE CASE-STUDY PAGE, TOP TO BOTTOM

- 1 **Hero shot:** the finished piece, big, first. Earn the scroll.
- 2 **The brief in a line:** the challenge and who it was for, from Lesson 8.1.
- 3 **Approach:** your strategy in a sentence or two: the key idea that solved it.
- 4 **Process:** 3–5 progress shots or a before/after that reveal how it was built.
- 5 **Details:** a few close crops of craft you're proud of (a clean cutout, the grade, the type).
- 6 **Result:** deliverables in context/mockups, plus any outcome you can claim.
- 7 **Reflection:** one honest paragraph on what you learned.

◆ STEP 1: ASSEMBLE THE ASSETS (IN CLASS)

Export a clean hero, 3–5 progress shots, a couple of detail crops, and a before/after. Place the hero and deliverables into mockups for context. No mockup of your own? Use `123-mockup-wall.jpg` (framed print), `123-mockup-phone.jpg` (social feed), or `123-mockup-sleeve.jpg` (album sleeve).

- 1 Draw a rectangle over the gray area where the art goes and **Convert to Smart Object**. Rotate or Warp it to match the surface if needed.
- 2 Double-click its thumbnail, place your hero in the tab that opens, save, and close.
- 3 Add the surface's shadows (Multiply) and highlights (Screen) above it, clipped. Swap art later with Layer ► Smart Objects ► Replace Contents.
- 4 Export `firstname-context.jpg` and `firstname-before-after.jpg` to the shared folder.

◆ STEP 2: WRITE THE STORY (IN CLASS)

Keep each beat tight, a few sentences. Lead with the outcome or the idea, not the tools: "A warm, editorial key art that reads on mobile" beats "I used Curves and Generative Fill."

Challenge (what needed solving):

Approach (your strategy and why):

Process (the three or four moments that show judgment; any generative assistance named here):

Result (the finished work in context):

Reflection (what worked, what you would change):

My 30-second spoken version (the answer to "tell me about this piece" in an interview):

◆ STEPS 3-4: LAY IT OUT, EXPORT & PUBLISH (AT HOME, WITHIN A WEEK)

- ◆ Build the page top to bottom: hero → brief → approach → process → details → result → reflection, in your Module 5 type system.
- ◆ Export web-sized images via **File ► Export ► Export As**, and a PDF via **File ► Save a Copy ► Photoshop PDF** (or **File ► Export ► Artboards to PDF** if you laid it out on several artboards). Automate the export if you built an Action (Module 6).
- ◆ Used Generative Fill or Expand? Say so in the process section. You can also attach Content Credentials on export (look for the option in Export As or the Content Credentials panel).
- ◆ Put it where people will see it: portfolio site, PDF, a post.

◆ PROJECT COMPLETION CHECKLIST (COURSE CAPSTONE)

- ◆ ☐ A big, clean hero shot leading the piece
- ◆ ☐ The five story beats written tight and outcome-first
- ◆ ☐ Honest before/after and 3–5 progress shots
- ◆ ☐ Deliverables shown in context via mockups
- ◆ ☐ Laid out with your type system, exported, and published
- ◆ ☐ Any generative-AI assistance disclosed honestly

◆ PRESENTATION & CRITIQUE

When I present (90 s): the hero in context, my brief in a line, the before/after, and my 30-second story. Then the room squints and says what they see first.

Feedback I received (what the room saw first · the glows · the grows):

The one grow I will act on:

When I review a classmate: read their brief first ("Which success criterion does it meet most clearly?"), then pick one rubric row and give one **glow** (what works, pointing at the pixels) and one **grow** ("I wonder if..."). Critique the work, not the person; judge against their brief, not your taste.

◆ RECIPE CARD: CASE STUDY

- 1 Lead with a big hero; earn the scroll
- 2 Five beats: challenge · approach · process · result · reflection
- 3 Show honest before/after + 3–5 progress shots
- 4 Present deliverables in context (mockups)
- 5 Lay out with your type system; export & publish

Mantra: show that you think like a professional, not just that you can push pixels.

◆ SELF-CHECK

Q1

What is the universal case-study arc, in order?

- A) Hero → tools → layers → export → thanks
- B) Before → after → mockup → price → contact
- C) Challenge → approach → process → result → reflection
- D) Moodboard → brief → build → grade → publish

Q2

What comes first on the case-study page?

- A) A big, clean hero shot of the finished piece
- B) A list of the tools and filters you used
- C) The moodboard
- D) The reflection paragraph

Q3

Which line leads with the outcome or the idea, as the lesson asks?

- A) "I used Curves and Generative Fill."
- B) "I spent twelve hours on this."
- C) "I tried lots of things until it worked."
- D) "A warm, editorial key art that reads on mobile."

Q4

Why present the final piece in a mockup?

- A) Mockups hide weak edges
- B) Context tells the reviewer what the thing is for and makes it look shipped rather than sketched
- C) Portfolio sites reject plain images
- D) It replaces the before/after

Q5

You laid the case study out in one document. How do you export it as a PDF?

- A) File ► Save a Copy ► Photoshop PDF
- B) File ► Export ► Quick Export as PNG
- C) File ► Scripts ► Image Processor
- D) Image ► Duplicate, then print to paper

Q6

Part of your piece used Generative Expand. What does the lesson say?

- A) Leave it out; reviewers only care about the result
- B) Rebuild the area by hand so there is nothing to disclose
- C) Say so in the process section; you can also attach Content Credentials on export
- D) Mention it only if a reviewer asks

◆ FINAL JOURNAL

Read back through your whole journal, from Module 1 to here. What can you do now that you couldn't when you started? Write that down: it is the real deliverable of this course.

◆ KEEP GROWING

- ◆ **Build your portfolio:** turn each capstone-quality project into a case study.
- ◆ **Practice deliberately:** pick one weak spot and drill it until it's a strength.
- ◆ **Study work you admire:** reverse-engineer the grade, the light, the type.
- ◆ **Seek critique:** ask someone in the field for honest feedback, then revise.
- ◆ **Specialize or broaden:** retouching, compositing, or design; or widen into motion and 3D.

Your very next project: something slightly beyond comfortable, briefed the way you learned in 8.1, finished and presented the way you learned in 8.3. Keep the **Recipes II** library open for the harder how-tos.

◆ NOTES

YOU FINISHED THE COURSE 🎉

You arrived able to operate Photoshop. Now you can cut anything out cleanly, grade for any screen or press, retouch a portrait invisibly, build a believable composite from nothing, run a consistent design system, automate volume work, combine many frames into one, and present it all like a pro, with the judgment to know which tool a problem needs and when to stop. This is not the end of your Photoshop journey; it's the point where it becomes yours.

Photoshop Intermediate: From Practitioner to Pro · Participant Workbook

Complete Participant Workbook · all 8 modules · v1.0 · © 2026 Ray de la Paz